SSI-GL: A Subpixel Spatial Index Guided Learning Framework for Self-Supervised Remote Sensing Image Subpixel Mapping

Junfei Zhong, Ke Wu, *Senior Member, IEEE*, Lina Xu, *Member, IEEE*, Yanfei Zhong, *Senior Member, IEEE*, Huize Liu, and Dandan Zhou

***Abstract*—Subpixel mapping (SPM) aims to generate high-resolution land cover maps from low-resolution imagery. However, end-to-end (ETE) deep learning (DL) methods are critically hampered by their reliance on extensive, paired training data, as high-resolution labels are costly and labor-intensive to acquire. This data scarcity severely limits the real-world application of current SPM algorithms. To overcome this challenge, we introduce the subpixel spatial index guided learning (SSI-GL) framework, a self-supervised learning (SSL) paradigm tailored for SPM. At its core, an innovative subpixel spatial index (SSI) mechanism guides the encoder to learn meaningful, spatially aware representations at the subpixel level during unsupervised pretraining. Subsequently, for the downstream mapping task, a semantic-constrained mapping head (SCMH) is employed, which leverages low-resolution semantic information to regularize the learning process and accelerate convergence. For the validation of our algorithm, four challenging public datasets were selected, encompassing RGB, hyperspectral, and multispectral imagery of spatially heterogeneous urban environments. Experiments included both simulated data and a real-world application, where 1-m land cover maps were generated from 10-m Sentinel-2 images. The experimental results show that integrating an SSL-pretrained encoder can reduce the need for labeled training data while maintaining strong performance. Across the conducted experiments, SSI-GL achieves the strongest overall results among the compared methods, suggesting its potential for generating fine-grained land cover maps under limited supervision. The source code is available at https://github.com/JunfeiZhong/SSI_GL**

***Index Terms*—Deep learning (DL), mixed pixel problem, remote sensing, self-supervised learning (SSL), subpixel mapping (SPM).**

I. Introduction

REMOTE sensing images, captured through various satellite-borne sensors, has found widespread applications across diverse fields, including lithological mapping [1], [2], scene classification [3], change detection [4], land use/cover mapping [5], [6], and spatiotemporal fusion [7], [8]. Within this spectrum, land use/cover mapping emerges as a pivotal methodology for characterizing surface configurations. The remote sensing imaging technology has undergone significant advancements, yielding a plethora of data options. While modern remote sensing technology has achieved remarkable progress, evidenced by high-resolution systems like ZY and GF series satellites, their operational effectiveness remains constrained by extended revisit cycles and limited swath widths, posing significant challenges for large-scale land cover identification projects. Paradoxically, while widely accessible datasets such as Landsat imagery offer temporal consistency and spatial coverage for such applications, their relatively coarse spatial resolution proves inadequate for detailed surface monitoring, particularly in urban environments characterized by complex land cover patterns [9]. This spatial resolution limitation inevitably leads to the prevalence of mixed pixels—a phenomenon where multiple land cover types coexist within a single pixel unit [10]. Conventional classification approaches, fundamentally constrained by their spectral analysis paradigms, fail to resolve the intricate spatial distribution patterns encapsulated in these mixed pixels. Consequently, existing methodologies can only generate generalized land cover representations at native sensor resolution, obscuring critical fine-grained spatial details essential for accurate surface feature characterization.

The spectral unmixing (SU) technique [11], [12], [13] has been proposed as a solution to mitigate the mixed pixel problem. By extracting the spectral signatures of carefully

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selected pure pixels (endmembers) corresponding to diverse land cover types, each mixed pixel will be decomposed into a combination of the constituent land cover types weighted by their corresponding fractional abundances, fundamentally differentiating itself from hard classification (HC) approaches that enforce single-label assignments to the mixed pixels. However, the SU-derived fractional abundance maps still maintain the original coarse spatial resolution, failing to capture the intricate intrapixel spatial variability. Consequently, the mixed pixel problem's detrimental impacts on fine land cover characterization persist [14], [15]. To reconstruct the potential spatial distribution patterns of various land cover types in mixed pixels, the subpixel mapping (SPM) [16] is proposed. By subdividing each mixed pixel into subpixel units according to a predetermined mapping scale and assigning land cover labels to subpixels, SPM produces land cover maps with higher spatial resolution, thereby improving the observable detail level.

Notably, the SPM process constitutes an ill-posed inverse problem [15], [17] due to its inherent information recovery mechanism through low-to-high-resolution (LR-HR) scale transformation. This mathematical indeterminacy manifests as multiple plausible solutions satisfying the same observational equations, necessitating constraining priors for generating rational and precise SPM outputs. These constraining priors are broadly categorized into two types based on data dependency: 1) endogenous constraints that leverage intrinsic spatial and/or spectral relationships within the target data through algorithms based on spatial dependency [18], [19], [20], [21], [22], [23] or distribution regularization [24], [25], [26], [27], [28], [29]; and 2) exogenous constraints that incorporate ancillary datasets via data fusion techniques [30], [31], [32] or learned representations [33], [34], [35]. However, these approaches face persistent challenges in being adapted to landscapes with complex spatial heterogeneity patterns, reducing dependency on the availability and quality of ancillary data, and resolving inherent uncertainties in prior quantification [14], [17]. These fundamental limitations currently restrict the operational robustness of SPM implementations across diverse terrestrial environments. Over recent years, the rapid advancement of deep learning (DL) technology has revolutionized the field of SPM. DL networks have demonstrated remarkable proficiency in learning sophisticated mapping functions from training data comprising paired LR images and their corresponding HR annotated maps, leveraging their powerful feature learning and representation capacities. These learned functions serve as improved constraining priors for SPM processes of the target data, offering superior flexibility and robustness compared to those traditional approaches, thereby achieving outstanding performance in reconstructing detailed HR land cover characteristics. Current DL-based SPM methods primarily follow two methodological trends. The predominant unmixing-then-mapping (UTM) approach [36], [37], [38], [39] employs sequential processing: first deriving abundance maps through SU and then generating SPM results through spatial mapping. Despite its popularity, this two-stage method suffers from critical limitations: 1) error propagation between stages leads to cumulative inaccuracies, as uncertainties in SU-derived abundance maps directly compromise subsequent mapping quality; and 2) spectral dimensionality inherently restricts classification capacity, posing significant limitations in multispectral image applications where the number of distinguishable land cover types cannot exceed the available spectral bands. These inherent drawbacks have catalyzed the emergence of end-to-end (ETE) DL-based SPM networks that synergistically integrate the unmixing and mapping processes [14]. By directly transforming LR images into refined SPM maps through unified neural networks, ETE frameworks [9], [14], [15], [40], [41], [42] effectively eliminate error accumulation risks while overcoming spectral dimensionality limitations. This integrated learning method demonstrates enhanced performance in handling complex land surface heterogeneity for SPM applications through joint optimization of spectral–spatial feature representation.

However, current SPM networks predominantly utilize supervised learning, a paradigm that heavily relies on substantial, accurately labeled, high-quality training datasets. Unfortunately, these ideal datasets are relatively scarce, particularly when juxtaposed with the immense volume of readily accessible remote sensing images. This data scarcity constitutes a major bottleneck for supervised learning methodologies [43], [44]. Furthermore, state-of-the-art supervised ETE SPM networks typically require grouped sets of LR images, LR pixel-level classification maps, and HR land cover maps [15], [17], [42]. Obtaining meticulously annotated HR land cover maps that are spatiotemporally consistent with LR data presents significant challenges, stemming from factors such as the high cost of manual annotation, potential annotation bias, temporal discrepancies in imaging, and geographic coordinate misalignments, among others. These combined issues render the acquisition of sufficient, high-quality labeled datasets exceptionally difficult. Therefore, an urgent need arises for SPM methods capable of learning more from less labels and less samples, as depicted in Fig. 1.

Self-supervised learning (SSL) has rapidly ascended as a remarkably promising paradigm within the computer vision community. It has achieved widespread acclaim for its capacity to efficiently acquire robust feature representations from copious unlabeled data [45], [46]. Applied across a spectrum of visual tasks [46], [47], [48], [49], [50] and increasingly extended to diverse domains of remote sensing image analysis in recent years [51], [52], [53], [54], SSL demonstrates significant versatility. Within an SSL framework, the network undergoes initial pretraining on an unlabeled dataset through a self-supervised pretext task. Subsequently, the learned representations are transferred to downstream tasks to address the target problem. In essence, SSL facilitates task completion by leveraging a substantial volume of unlabeled data while requiring only a minimal quantity of labeled data. However, the direct application of existing SSL methodologies to SPM is fundamentally challenged by a mismatch in spatial scale. Mainstream SSL frameworks, conceived for image-level tasks, learn representations by comparing globally pooled feature vectors [55], [56], [57], [58]. This process intentionally discards the fine-grained spatial information that is the very essence of SPM. Alternative dense SSL methods that endeavor

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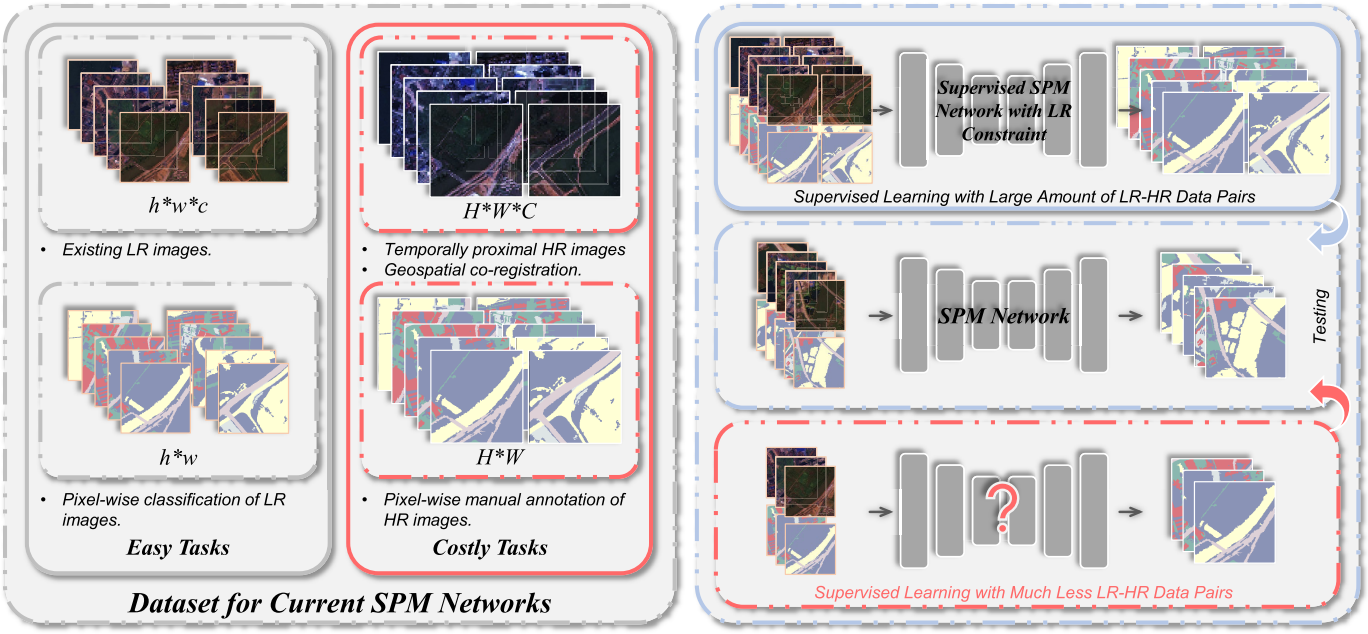


Fig. 1. While supervised SPM methods heavily rely on large amounts of labeled HR data, a paradigm with less supervision is urgently needed for SPM tasks.

to capture pixel-level features often approximate local feature matching between augmented views [59], [60], [61], lacking the spatial precision required for SPM. The core objective of SPM, in contrast, is to resolve the spatial heterogeneity within a single mixed pixel. Consequently, an SSL approach that operates at the image or interpixel level cannot provide suitable representations for what is inherently an intrapixel problem, which means that a more refined subpixel-level feature monitoring and matching methodology is imperative to effectively integrate SSL into the SPM domain for facilitating the capture of features with a finer level of granularity during the crucial pretraining phase.

In this study, we present a subpixel spatial index guided learning (SSI-GL) framework for self-supervised remote sensing image SPM, in which a novel subpixel spatial index (SSI) mechanism is introduced to guide subpixel-level representation learning during pretraining. The main contributions of this work are summarized as follows.

1) We introduce SSL into remote sensing image SPM and propose a dedicated SSL-based framework, termed SSI-GL, to alleviate the strong dependence of existing DL-based SPM methods on scarce high-resolution annotations. By leveraging large-scale unlabeled data during pretraining, SSI-GL enables the encoder to learn robust and transferable representations for label-efficient SPM.
2) We propose the SSI mechanism for self-supervised subpixel-level representation learning. Specifically, SSI employs index masks to establish explicit spatial correspondences between two augmented views and track matched pixel positions on feature maps with the target spatial resolution. This design enforces consistency at the subpixel level and allows the encoder to learn discriminative subpixel-invariant representations tailored to SPM.
3) We design a semantic-constrained mapping head (SCMH) to efficiently adapt the pretrained encoder to the downstream SPM task. By incorporating low-resolution semantic constraints, SCMH accelerates convergence and further improves mapping performance.
4) Extensive experiments on three simulated datasets and one real-world dataset demonstrate the effectiveness and generality of the proposed method. Across the conducted experiments, SSI-GL achieves the strongest overall results among the compared supervised SPM methods and SSL baselines, especially under limited-label settings. Moreover, experiments on real Sentinel-2 imagery indicate the practical potential of integrating SSL into SPM for reducing the reliance on costly manual high-resolution annotation.

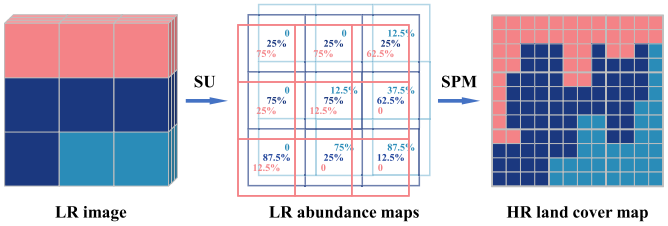


Fig. 2. SU and SPM processes of traditional SPM methods.

II. Background

A. DL-Based SPM Formulation

Traditional SPM methods partition the mixed pixels into subpixels according to a predefined scale, followed by label assignment to the subpixels based on the abundance maps

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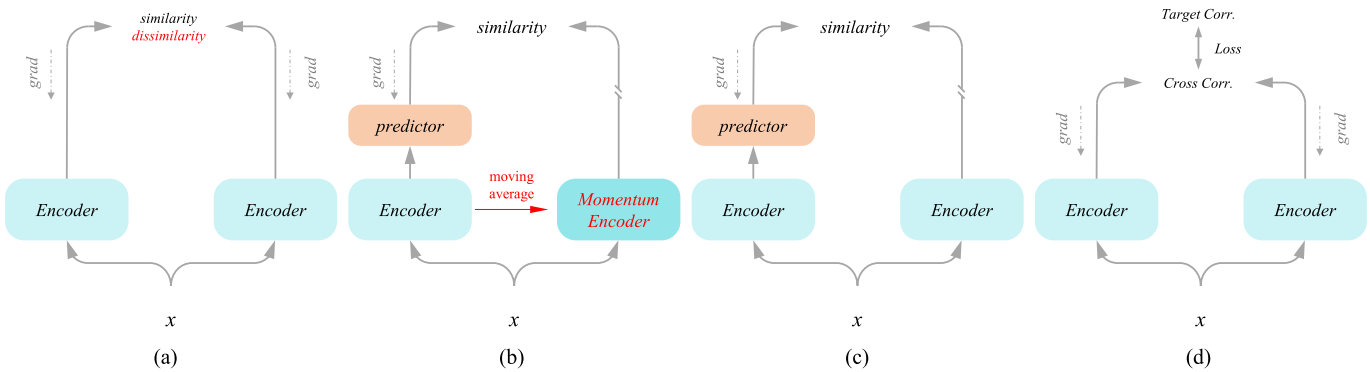


Fig. 3. Various SSL networks. (a) SimCLR, (b) BYOL, (c) SimSiam, and (d) Barlow Twins.

derived from SU, as shown in Fig. 2. Furthermore, the mathematical framework of DL-based SPM should be grounded in the SU process, where a given LR image $\boldsymbol{X} \in \mathbb{R}^{h\times w\times c}$ with a height of h, width of w, and c channels can be jointly represented by the abundance matrix $\boldsymbol{A} \in \mathbb{R}^{h\times w\times NC}$ of NC land cover types, the mixed spectral matrix $\boldsymbol{S} \in \mathbb{R}^{NC\times c}$ and the bias matrix $\boldsymbol{b} \in \mathbb{R}^{h\times w\times c}$ through a mixing function $f_{\mathrm{M}}(\cdot)$

$$\boldsymbol{X} = f_{\mathrm{M}}(\boldsymbol{A}, \boldsymbol{S}, \boldsymbol{b}). \tag{1}$$

In this context, the LR abundance matrix $\boldsymbol{A}$ can be formally formulated as a degraded representation of the one-hot encoded HR land cover map $\boldsymbol{Y} \in \mathbb{R}^{H\times W\times NC}$ through the degradation function $f_{\mathrm{D}}(\cdot)$

$$\boldsymbol{A} = f_{\mathrm{D}}(\boldsymbol{Y}). \tag{2}$$

Drawing from (1) and (2), a functional mapping is established between $\boldsymbol{X}$ and $\boldsymbol{Y}$

$$\boldsymbol{X} = f_{\mathrm{M}}(f_{\mathrm{D}}(\boldsymbol{Y}), \boldsymbol{S}, \boldsymbol{b}). \tag{3}$$

Given that SPM $g_{\mathrm{SPM}}(\cdot)$ generates HR land cover maps from LR images, the underlying mapping is formulated as

$$\boldsymbol{Y} = g_{\mathrm{SPM}}(\boldsymbol{X}, \boldsymbol{S}, \boldsymbol{b}). \tag{4}$$

Ultimately, the supervised DL-based SPM network $\mathcal{N}(\boldsymbol{X};\boldsymbol{\theta})$ takes an input $\boldsymbol{X}$ and predicts a corresponding HR probability map $\hat{\boldsymbol{Y}} \in \mathbb{R}^{H\times W\times NC}$ and the network parameters $\boldsymbol{\theta}$ are updated through backpropagation driven by the loss function $\mathcal{L}(\hat{\boldsymbol{Y}}, \boldsymbol{Y})$ which is minimized during the training, thereby ensuring that the mapping process converges to the mathematical relationship described in (4).

B. Self-Supervised Learning

SSL aims to leverage the data itself as a supervised signal, enabling neural networks to learn generic, transferable feature representations from the intrinsic structures inherent in unlabeled data through various pretext tasks. Currently, contrastive SSL stands as one of the predominant methodologies. The core idea behind these methods, such as SimCLR [Fig. 3(a)] [55], is to maximize the similarity between features of positive sample pairs while simultaneously minimizing the similarity between features of negative sample pairs. This process encourages the network to learn invariant representations robust to data augmentation operations. Building upon this, subsequent innovations have streamlined the contrastive self-supervised paradigm by eliminating the reliance on negative sampling, thereby simplifying the learning objective. Prominent frameworks such as BYOL [Fig. 3(b)] [56] and SimSiam [Fig. 3(c)] [57] focus solely on aligning positive sample pairs. To effectively circumvent representation collapse in this negative-free setting, they employ asymmetric network architectures and specific training dynamics, including momentum encoders, predictors, and stop-gradient operations. Furthermore, methods based on the principle of redundancy reduction, such as Barlow Twins (BT) [Fig. 3(d)] [58], minimize the difference between the cross-correlation matrix of feature vectors output by twin networks and an identity matrix. This approach simultaneously achieves invariance learning and redundancy reduction, thereby learning invariant features with low redundancy. These advancements have significantly propelled the development of unsupervised feature learning and demonstrated superior performance across various downstream tasks.

III. Methodology

The proposed SSI-GL framework simultaneously focuses on two augmented views of features at both image level and subpixel level, where the correlation losses of features are minimized. Furthermore, the SCMH is developed to enhance the downstream SPM performance. This section is dedicated to a comprehensive elaboration on the self-supervised pretraining and the downstream process.

A. SSI-GL Framework Overview

Our work adapts the powerful principle of redundancy reduction from SSL to the specific challenges of the SPM domain. However, a direct application of such frameworks is fundamentally suboptimal. They are designed to learn image-level invariance by aggregating spatial features into a single global vector, a process that discards the intrapixel spatial information essential for SPM. To overcome this limitation, we introduce the SSI mechanism, which seeks to foster precise spatial positioning and consistency monitoring of subpixel feature from two augmented views during the pretraining stage. The overall architecture of the proposed SSI-GL framework is illustrated in Fig. 4. Specifically, two augmented views

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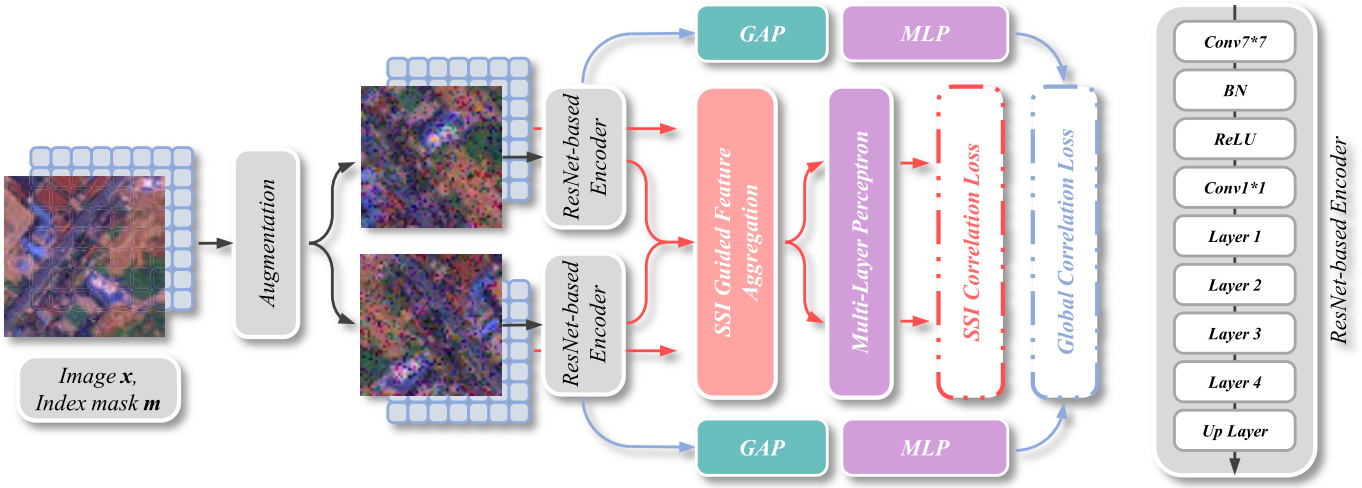


Fig. 4. Overall architecture of the proposed SSI-GL framework.

of an image are processed by a shared encoder to generate a pair of feature representations. This feature pair is then fed into two parallel branches: an SSI correlation branch and a global correlation branch. Within the SSI correlation branch, the encoder features first pass through an SSI-guided feature aggregation (SGFA) to form a pair of aggregated linear features. Subsequently, a weight-sharing multilayer perceptron (MLP) transforms this aggregated pair into a pair of projected features for the SSI correlation loss calculation. Concurrently, in the global correlation branch, the encoder feature pair first undergoes global average pooling (GAP) to create a pair of global linear features. These features are then processed by another MLP to produce a projected feature pair, upon which the global correlation loss is computed. Finally, the SSI correlation loss and the global correlation loss are dynamically weighted and summed according to their respective magnitudes.

1) Data Augmentation: Our data augmentation pipeline is specifically tailored for the SPM task and consists of two categories of transformations.

1) *Geometric Augmentations:* Basic spatial transformations, including random cropping, flipping, and rotations, are applied identically to both the image views and their corresponding index masks. Since the index mask has the same spatial size as the input image, we first stack them together before each geometric augmentation and separate them afterward, ensuring that the two always share the exact same crop coordinates, flipping states, and rotation angles.
2) *Masking-based Augmentation:* We then apply random binary masking to the base augmented views, where a fraction of pixels in each augmented view is randomly set to zero across all spectral channels. This encourages the model to learn robust representations from partial information. Critically, we intentionally omit all photometric augmentations, such as color jittering, brightness adjustments, and Gaussian blurring. The SPM task is sensitive to the original spectral signatures of land covers. Preserving this spectral integrity is paramount, and altering pixel values would interfere with the model's ability to learn the fine-grained features necessary for accurate SPM.

2) SSI Correlation Loss: In the proposed framework, the subpixel-level feature representation is emphasized by the SSI mechanism, thereby facilitating self-supervised training of the network. The calculation of SSI correlation loss entails localizing each mixed pixel in the two augmented views to its corresponding position in the original image using an index mask, and subsequently tracking and pairing the subpixel-level feature vectors generated from the mixed pixels with identical indices. As illustrated in Fig. 5, the subpixel feature vectors within each nonoverlapping $S \times S$ region on the HR output features of the pretrained network will be summed to establish a one-to-one correspondence between the accumulated feature vectors and their corresponding mixed pixels. Subsequently, guided by the index masks, feature vector pairs with identical indices are selected from the accumulated feature pair and further aggregated, yielding the SSI-guided aggregated linear features, which is then transformed by a projector into a pair of linear projected features for computing the cross-correlation matrix-based SSI correlation loss.

3) Balanced Loss of the SSI and Global Correlation Loss: The SSI-GL framework takes as input two augmented views consisting of images and masks—$[\boldsymbol{x}_1, \boldsymbol{m}_1]$ and $[\boldsymbol{x}_2, \boldsymbol{m}_2]$, in which $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are processed by a modified ResNet-based encoder $E(\cdot)$. The SSI correlation branch applies SGFA $F_{\mathrm{SGFA}}(\cdot)$ to the features and masks, followed by an MLP projector $H_{\mathrm{SSI}}(\cdot)$ to yield a feature pair for SSI correlation loss $\mathcal{L}_{\mathrm{SSI}}(\cdot)$. Meanwhile, the encoder features passed into the global correlation branch are directly transformed into linear features via GAP $F_{\mathrm{GAP}}(\cdot)$ and then processed by an MLP projector $H_{\mathrm{G}}(\cdot)$ to generate global features for computing the global correlation loss $\mathcal{L}_{\mathrm{G}}(\cdot)$. Denoting the output features of the SSI correlation branch as $\boldsymbol{z}_1^{\mathrm{SSI}} \triangleq H_{\mathrm{SSI}}(F_{\mathrm{SGFA}}(E(\boldsymbol{x}_1), \boldsymbol{m}_1))$ and $\boldsymbol{z}_2^{\mathrm{SSI}} \triangleq H_{\mathrm{SSI}}(F_{\mathrm{SGFA}}(E(\boldsymbol{x}_2), \boldsymbol{m}_2))$, the output features of the global correlation branch as $\boldsymbol{z}_1^{\mathrm{G}} \triangleq H_{\mathrm{G}}(F_{\mathrm{GAP}}(E(\boldsymbol{x}_1)))$ and

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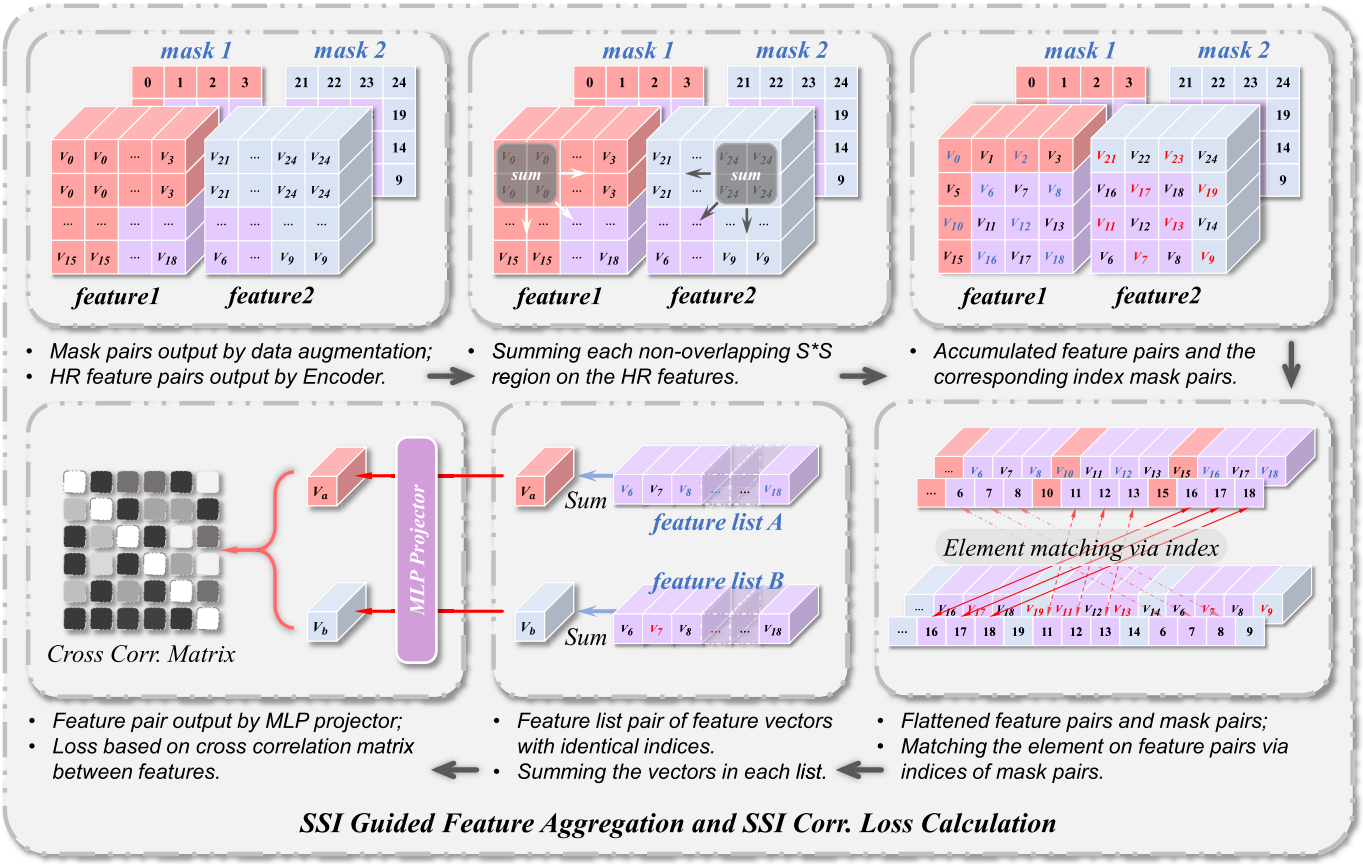


Fig. 5. Flowchart of SSI correlation loss calculation.

$z_2^{\mathrm{G}} \triangleq H_{\mathrm{G}}(F_{\mathrm{GAP}}(E(\boldsymbol{x}_2)))$, we aim to minimize the correlation loss $\mathcal{L}_{\mathrm{Corr.}}(\cdot)$ between z_1^{SSI} and z_2^{SSI}, z_1^{G} and z_2^{G}

$$\mathcal{L}_{\mathrm{Corr.}}(z_1, z_2) \triangleq \frac{\sum_i (1 - \mathrm{C}_{ii}(z_1, z_2))^2 + \lambda \sum_i \sum_{i \neq j} \mathrm{C}_{ij}(z_1, z_2)^2}{d} \tag{5}$$

$$\mathrm{C}_{ij}(z_1, z_2)^2 \triangleq \frac{\sum_b z_{1,b,i} z_{2,b,j}}{\sqrt{\sum_b (z_{1,b,i})^2} \sqrt{\sum_b (z_{2,b,j})^2}} \tag{6}$$

where d is the dimensionality of the feature vectors z_1 and z_2, b represents the batch size, and λ is a parameter with a default value of $5\mathrm{e}^{-3}$ for balancing the two terms. To effectively facilitate the learning of unsupervised representations at both the subpixel and image levels during the pretraining stage, we introduce a balanced loss function. Specifically, this function operates by dynamically weighting the contribution of each branch's correlation loss in accordance with its magnitude

$$\mathcal{L} = w_1 \mathcal{L}_{\mathrm{SSI}}\left(z_1^{\mathrm{SSI}}, z_2^{\mathrm{SSI}}\right) + w_2 \mathcal{L}_{\mathrm{G}}\left(z_1^{\mathrm{G}}, z_2^{\mathrm{G}}\right) \tag{7}$$

$$w_1 = \frac{exp\left(\mathcal{L}_{\mathrm{SSI}}\left(z_1^{\mathrm{SSI}}, z_2^{\mathrm{SSI}}\right)\right)}{exp\left(\mathcal{L}_{\mathrm{SSI}}\left(z_1^{\mathrm{SSI}}, z_2^{\mathrm{SSI}}\right)\right) + exp\left(\mathcal{L}_{\mathrm{G}}\left(z_1^{\mathrm{G}}, z_2^{\mathrm{G}}\right)\right)} \tag{8}$$

$$w_2 = \frac{exp\left(\mathcal{L}_{\mathrm{G}}\left(z_1^{\mathrm{G}}, z_2^{\mathrm{G}}\right)\right)}{exp\left(\mathcal{L}_{\mathrm{SSI}}\left(z_1^{\mathrm{SSI}}, z_2^{\mathrm{SSI}}\right)\right) + exp\left(\mathcal{L}_{\mathrm{G}}\left(z_1^{\mathrm{G}}, z_2^{\mathrm{G}}\right)\right)}. \tag{9}$$

This dynamic weighting scheme automatically balances the contributions of the two branches. By normalizing the loss magnitudes through a softmax function, it allows the network to dynamically focus on the more challenging task (i.e., the branch with the higher current loss) during training, preventing either loss from dominating the gradient updates.

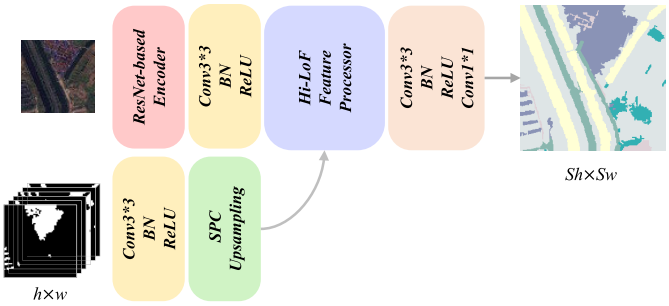


Fig. 6. Architecture of the SCMH for the downstream SPM task.

B. Downstream SPM With SCMH

To adapt the powerful representations learned by the pretrained encoder for the final SPM task, we introduce the SCMH, as depicted in Fig. 6. This head is designed to translate the encoder's high-resolution features into the land cover map, leveraging low-resolution semantic information as a guiding constraint to accelerate training convergence. The SCMH mainly consists of a high- and low-frequency (Hi-LoF) feature processor and a classifier with two convolution layers. Specifically, the Hi-LoF feature processor

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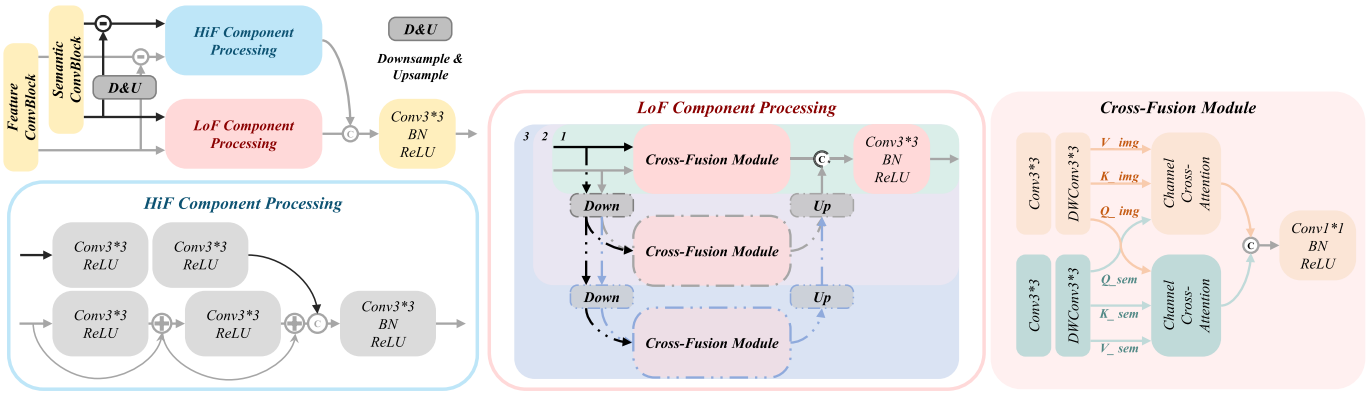


Fig. 7. Architecture of Hi-LoF Feature Processor.

leverages the upsampled LR semantic information to constrain and modulate the HiF and LoF components of the network features, and the classifier maps the constrained outputs to categorical features of land covers.

As illustrated in Fig. 7, the Hi-LoF Feature Processor calculates the difference between the input features and those that have undergone average-pooling downsampling and bilinear interpolation upsampling, yielding the HiF components of the image features and semantic features. These HiF components are then fed into the HiF component processing branch. Concurrently, the LoF component processing branch implements a hierarchical processing approach, whereby features are processed at varying spatial resolutions. Specifically, the image features and semantic features are directly input into a cross-fusion module containing dual-channel cross-attention (CCA) to achieve semantic constraints and feature interactions at the target spatial resolution. Then, the image features and semantic features are downsampled to the medium-resolution (MR) background to create an additional branch, where the features are fused and then bilinearly interpolated to be concatenated with the fused features from the previous branch. The same process is repeated when image features and semantic features are downsampled to the LR background.

In the CCA computation involved in the aforementioned process, the image features $\boldsymbol{F}_{\text{img}} \in \mathbb{R}^{C\times H\times W}$ and semantic features $\boldsymbol{F}_{\text{sem}} \in \mathbb{R}^{C\times H\times W}$ are first transformed to their respective queries $\boldsymbol{Q}_{\text{img}}, \boldsymbol{Q}_{\text{sem}} \in \mathbb{R}^{C\times HW}$, keys $\boldsymbol{K}_{\text{img}}, \boldsymbol{K}_{\text{sem}} \in \mathbb{R}^{C\times HW}$, and values $\boldsymbol{V}_{\text{img}}, \boldsymbol{V}_{\text{sem}} \in \mathbb{R}^{C\times HW}$. Subsequently, the cross-attention maps along the channel dimension are computed, which are then used to weight the corresponding values

$$\boldsymbol{F}_1 = softmax\left(\frac{\boldsymbol{Q}_{\text{img}}\boldsymbol{K}_{\text{sem}}^{\text{T}}}{\sqrt{C}}\right)\boldsymbol{V}_{\text{sem}} \tag{10}$$

$$\boldsymbol{F}_2 = softmax\left(\frac{\boldsymbol{Q}_{\text{sem}}\boldsymbol{K}_{\text{img}}^{\text{T}}}{\sqrt{C}}\right)\boldsymbol{V}_{\text{img}}. \tag{11}$$

IV. Experiments

A. Datasets

Given that the training process of the downstream SPM task still requires registered LR remote sensing images and HR labels, and considering the scarcity of relevant datasets in the current community, four publicly available land cover semantic segmentation datasets, namely WHDLD [62], EarthVQA [63], WHU-H^{2}SR [6], and WHU-MT [6], were selected for experimental validation of the proposed method. For each dataset, the corresponding preprocessing, training, and evaluation procedures were carried out independently.

1) *WHDLD dataset* provides 4940 HR RGB images with a spatial resolution of 2 m and a size of 256 × 256 pixels, located in the urban area of Wuhan, China, along with corresponding land cover maps categorized into six classes: building, road, pavement, vegetation, bare soil, and water. During the data preprocessing, the HR images were downsampled by a factor of 4 to generate simulated LR images with a spatial resolution of 8 m and a size of 64 × 64 pixels, which were used as input images for both the SSL pretraining and the downstream SPM task. Meanwhile, the HR land cover maps were downsampled by selecting the majority class of each nonoverlapping 4 × 4 region to provide LR semantic information for the downstream SPM task.
2) *EarthVQA dataset* offers 4190 HR RGB images with a spatial resolution of 0.3 m and a size of 1024 × 1024 pixels, accompanied by land cover semantic annotations categorized into eight classes: background, building, road, water, barren, forest, agriculture, and playground. Given that land covers at an observational scale of this level do not exhibit significant distributional differences caused by mixed pixels, the original images and land cover maps were downsampled to 1.2 m, resulting in 4190 pairs of 256 × 256 sized image patches as HR data. Subsequently, the data were processed in the same way as the WHDLD dataset to generate simulated LR images with a spatial resolution of 4.8 m and LR semantic information.
3) *WHU-H^{2}SR dataset* contains 2531 249-band H^{2}SR hyperspectral images with a spatial resolution of 1 m and a size of 300 × 300 pixels, along with corresponding land cover labels, collected in September 2020. This dataset comprises nine categories: paddy field, dry farmland, forest land, grassland, building, highway, greenhouse, water body, and background. During the

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data preprocessing, all image data were first cropped to 256 × 256 pixels to ensure consistency with other datasets and then underwent the same processing as other datasets to generate LR hyperspectral images and semantic information with a spatial resolution of 4 m.

4) *WHU-MT dataset* contains 2531 image sets, where each set includes a time series of 10-band, 10-m resolution, 30×30 pixels Sentinel-2 images acquired in every month of 2020, complete with 10-m resolution land cover maps. Each LR image is spatially co-registered with an identically numbered HR image from the WHU-H^{2}SR dataset strictly, and both share an identical nine-class system (paddy field, dry farmland, etc.). To ensure temporal alignment with the HR data's acquisition time, the Sentinel-2 image from September 2020 was selected from the multitemporal collection as the experimental data. During the data preprocessing stage, both the LR images and their corresponding maps were upsampled to 75 × 75 pixels via nearest neighbor interpolation, serving as the LR input and LR semantic information, respectively. This step was performed to better suit the SPM process while preserving the original 10-m observation level of the LR data. The 1-m HR ground truth (GT) is sourced from the corresponding WHU-H^{2}SR label. Crucially, the evaluation on the WHU-MT dataset was based entirely on genuine satellite data, avoiding synthetic simulations and thus establishing a more credible and realistic benchmark for performance assessment.

B. Implementation Details

The proposed SSI-GL framework and SCMH module were implemented in PyTorch, with all experiments conducted on a server equipped with a 24-GB NVIDIA GeForce RTX 3090 GPU. We adopted ResNet50 as the encoder backbone, modifying its architecture significantly to preserve full spatial resolution for dense subpixel prediction. Specifically, all downsampling operations were removed: 1) the stride of the initial 7 × 7 convolution (conv1) was changed from 2 to 1; 2) the subsequent MaxPool2d layer was replaced with a 1 × 1 convolutional layer using a stride of 1; and 3) the convolutions with the stride of 2, which are normally present at the beginning of layer2, layer3, and layer4 to perform downsampling, were all set to use a stride of 1. These modifications ensure that the essential spatial information for SPM is maintained throughout the forward pass. Additionally, the final GAP and fully connected layers were removed, and upsampling layers were appended to the encoder output to ensure that the resulting feature maps match the target spatial resolution, as depicted in Fig. 4. To accommodate datasets with different spectral dimensionalities, only the number of input channels in the encoder's first convolutional layer is adjusted according to each dataset, while all remaining architectural components are kept identical across experiments to ensure fair and controlled comparisons.

Our training protocol consisted of two stages: self-supervised pretraining and supervised downstream SPM. For the pretraining stage, the encoder was trained on all LR images from each dataset. We utilized the Adam optimizer (weight decay = 0) and a cosine annealing scheduler that dynamically adjusted the learning rate between $1e^{-6}$ and $1e^{-3}$. After 200 epochs with a batch size of 4, the model parameters yielding the minimum pretraining loss were saved for the subsequent downstream tasks.

For the supervised training stage on downstream SPM tasks, we utilized a minimal amount of labeled data. The effectiveness of the pretrained encoder was validated using two distinct strategies: freezing its weights (freezing, FZ) and fine-tuning them (fine-tuning, FT). For this stage, network optimization was performed using an SGD optimizer with a momentum of 0.9 and zero weight decay, again coupled with a cosine annealing scheduler for dynamically adjusting the learning rate from $1e^{-6}$ to $1e^{-3}$. To provide a comprehensive comparison, we also established a crucial baseline by training an identical network architecture from scratch [i.e., with random initialization (RI)] directly on the supervised task.

To benchmark our method, we selected a comprehensive set of comparative models based on their relevance and performance. These include four advanced information-constrained ETE supervised SPM networks (SIMNet [17], SCNet [41], SFRNet-MLFF [15], and MSMCNet [42]) and three widely used negative-free SSL methods (BYOL [56], SimSiam [57], and BT [58]). To ensure a fair comparison, these self-supervised methods employed an identical encoder architecture to that of our SSI-GL framework, and all were connected to the SCMH in the downstream SPM task. The best-performing results from all methods on the downstream tasks were used for the final comparative analysis.

To quantitatively evaluate the SPM performance of the SSI-GL framework and SCMH, we selected overall accuracy (OA), Kappa coefficient, mean intersection over union (mIoU), and the $F1$-score for each class as evaluation metrics. These four metrics are calculated based on the confusion matrix shown in Table I.

TABLE I
CONFUSION MATRIX

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

OA is the most commonly used evaluation metric, representing the proportion of correctly predicted samples to the total number of samples

$$\mathrm{OA} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{12}$$

The Kappa coefficient is a metric used to measure the agreement between the classifier's predictions and the true labels, taking into account the effect of chance. The Kappa coefficient ranges from −1 to 1, where 1 indicates perfect agreement, 0 represents random guessing, and −1 denotes complete disagreement

$$\kappa = \frac{(p_o - p_e)}{(1 - p_e)} \tag{13}$$

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TABLE II
QUANTITATIVE COMPARISON ON WHDLD DATASET

	SCNet		SIMNet		SFRNet-MLFF		MSMCNet		RI	BYOL	SimSiam	BT	SSI-GL (Ours)	
	1%	10%	1%	10%	1%	10%	1%	10%	1%	1%	1%	1%	FZ (1%)	FT (1%)
OA (%)	44.43	80.50	47.42	86.63	73.79	87.04	58.20	89.85	85.41	89.41	93.40	94.29	**94.81**	**94.82**
$\kappa(\times 100)$	-	70.79	12.54	80.33	61.08	81.11	38.26	85.13	78.43	84.40	90.48	91.78	**92.54**	**92.56**
mIoU (%)	7.41	41.53	12.73	55.49	34.53	52.52	23.79	64.91	57.06	67.17	81.73	84.63	**86.32**	**86.36**
building	-	0.7099	0.1317	0.8462	0.0172	0.7135	0.4536	0.8665	0.6682	0.8279	0.8942	0.9010	**0.9104**	**0.9107**
road	-	-	-	0.4549	0.1209	-	-	0.6349	0.6767	0.7399	0.8941	0.9148	**0.9265**	**0.9275**
pavement	-	0.5139	0.3353	0.6730	0.4923	0.8653	0.2386	0.8334	0.7573	0.7875	0.8655	0.8817	**0.8905**	**0.8901**
vegetation	0.6117	0.9153	0.6493	0.9312	0.8908	0.9397	0.6821	0.9464	0.7573	0.9256	0.9468	0.9534	**0.9575**	**0.9576**
bare soil	-	0.0137	-	0.0111	0.0646	-	-	0.1760	0.1837	0.4300	0.8002	0.8565	**0.8846**	**0.8850**
water	-	0.8465	-	0.9525	0.9071	0.9691	0.6178	0.9656	0.9183	0.9683	0.9748	0.9807	**0.9813**	**0.9815**

$$p_o = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

$$p_e = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{(TP + TN + FP + FN)^2} \tag{15}$$

where mIoU represents the average IoU across all classes. IoU is a metric used to evaluate the performance of image segmentation algorithms, defined as the ratio of the intersection to the union between predicted results and GT

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{16}$$

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \text{IoU}_i. \tag{17}$$

The $F1$-score is the harmonic mean of precision (Pre.) and recall (Re.), providing a comprehensive measure of the classifier's performance

$$F1 = \frac{2 \times Pre. \times Re.}{Pre. + Re.} \tag{18}$$

$$Pre. = \frac{TP}{TP + FP} \tag{19}$$

$$Re. = \frac{TP}{TP + FN}. \tag{20}$$

C. Experiment 1: WHDLD Dataset

To evaluate the effectiveness of SSI-GL and SCMH under extremely limited supervision, 1% of the labeled samples were randomly selected for training, 10% for validation, and the remaining samples for testing. Considering the stronger label dependence of supervised methods, we further trained four supervised comparative methods (SCNet, SIMNet, SFRNet-MLFF, and MSMCNet) with 10% labeled data to provide a broader comparison across different supervision levels. The quantitative results are reported in Table II.

Under the current data split and evaluation setting, SSI-GL achieves the best overall performance on the WHDLD dataset when the pretrained encoder is fine-tuned, with OA (%), $\kappa(\times 100)$, and mIoU (%) reaching 94.82%, 92.56, and 86.36%, respectively. The FZ strategy yields very similar results, suggesting that the pretrained encoder transfers effectively to the downstream SPM task on this dataset. Compared with generic SSL methods (BYOL, SimSiam, and BT), SSI-GL provides the strongest overall results in this experiment, which is consistent with the expected benefit of subpixel-aware pretraining. In contrast, RI and the four supervised models show clear performance limitations at the 1% label ratio. Although increasing the proportion of labeled data to 10% substantially improves the supervised baselines, their performance on WHDLD still remains below the SSL-based results reported in Table II. Overall, these observations suggest that SSL-based pretraining is a promising strategy for improving SPM performance under limited supervision.

For a qualitative comparison, Fig. 8 visualizes the results of four test images from the WHDLD dataset. As supervised methods show limited reliability under 1% labels, their visualizations present results trained on 10% labels. The comparison reveals that SCNet broadly captures large-scale spatial distributions but yields coarse details and numerous misclassifications. SIMNet shows better accuracy but is compromised by significant noise. SFRNet-MLFF and MSMCNet provide improved visual results, yet they still suffer from noticeable misclassification issues under limited training data.

When SSL-pretrained encoders are integrated with SCMH, all SSL variants show clear improvements over supervised counterparts in overall SPM quality. Since these SSL methods share the same downstream architecture and training protocol, the observed performance gaps mainly reflect differences in the quality of the learned representations. BYOL improves global structure recovery but still presents discontinuities and noticeable noise in fine details. SimSiam reconstructs major land-cover patterns more coherently, yet remains less reliable for thin linear and point-like structures. Barlow Twins further improves spatial continuity, although subtle boundaries in categories such as buildings and pavements are still not fully preserved. In contrast, SSI-GL provides the most faithful visual reconstruction, showing closer agreement with the GT in both scene-level spatial layout and local boundary details.

D. Experiment 2: EarthVQA Dataset

Following the same preprocessing pipeline and split protocol as in the WHDLD experiment, we randomly sampled 1% of the annotated data for training, 10% for validation, and used the remaining samples for testing. In addition, four supervised counterparts were further trained with 10% labeled data to provide a broader comparison. The quantitative results are reported in Table III. Because the playground class contains

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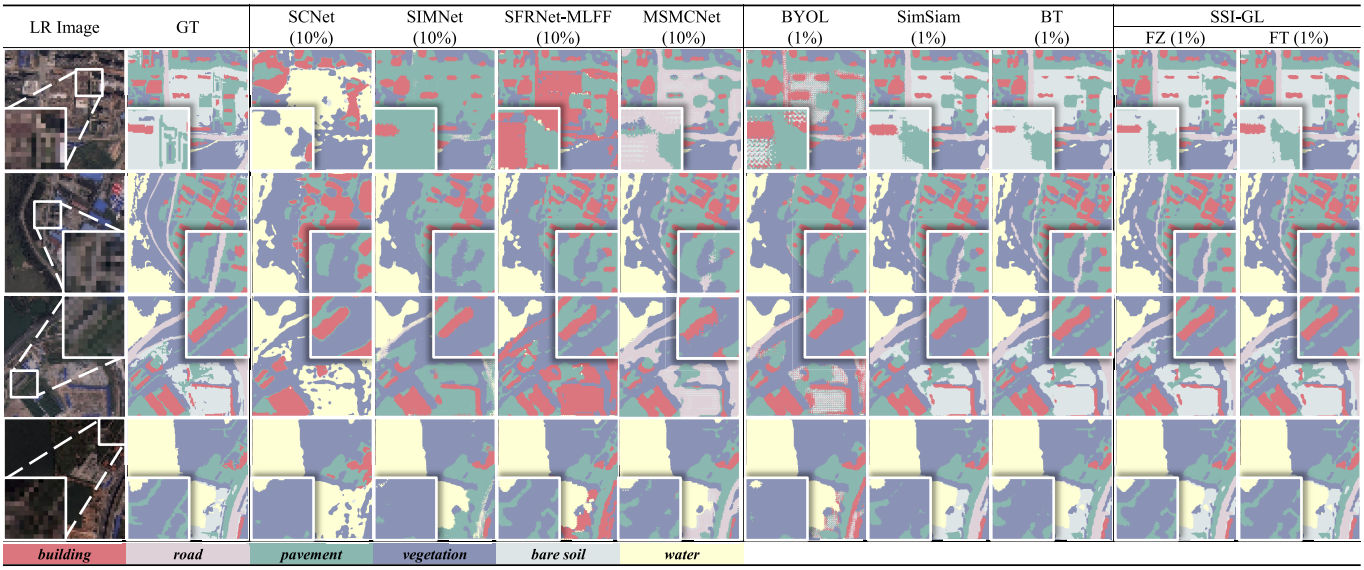


Fig. 8. Visual comparison of SPM results on the WHDLD dataset.

TABLE III
QUANTITATIVE COMPARISON ON THE EARTHVQA DATASET

	SCNet		SIMNet		SFRNet-MLFF		MSMCNet		RI	BYOL	SimSiam	BT	SSI-GL (Ours)	
	1%	10%	1%	10%	1%	10%	1%	10%	1%	1%	1%	1%	FZ (1%)	FT (1%)
OA (%)	46.88	78.78	36.72	74.51	71.07	91.18	30.66	86.24	86.03	91.10	92.92	94.77	**96.00**	**96.01**
$\kappa(\times 100)$	20.24	72.81	10.55	65.77	59.54	88.35	-	82.06	81.45	88.46	90.75	93.20	**94.80**	**94.81**
mIoU (%)	11.33	44.01	8.37	34.77	28.19	65.49	4.50	57.17	56.78	73.23	72.99	77.12	**85.58**	**85.80**
background	0.5595	0.8234	0.4942	0.8676	0.7566	0.9022	0.4672	0.8756	0.8844	0.8905	0.9233	0.9402	**0.9521**	**0.9519**
building	-	0.7152	-	0.4374	-	0.9225	-	0.8491	0.6533	0.9020	0.8968	0.9224	**0.9367**	**0.9361**
road	-	0.1935	-	0.0514	0.0382	0.8797	-	0.5834	0.2615	0.8344	0.7653	0.8770	**0.9054**	**0.9059**
water	-	0.6400	-	0.0222	-	0.9613	-	0.7922	0.8264	0.9484	0.9138	0.9551	**0.9688**	**0.9619**
barren	-	-	-	0.4690	-	0.0065	-	0.3572	0.8169	0.9337	0.9334	0.9294	**0.9619**	**0.9619**
forest	-	0.8528	0.1364	0.6749	0.8128	0.9442	-	0.8820	0.8169	0.9471	0.9373	0.9547	**0.9593**	**0.9618**
agriculture	0.6756	0.9592	0.4124	0.9260	0.9642	0.9731	0.0953	0.9509	0.8169	0.9053	0.9692	0.9706	**0.9800**	**0.9814**
playground	-	-	-	-	-	-	-	-	-	-	-	-	-	-

too few samples for statistically stable estimation, its $F1$-score is not included.

As shown in Table III, SSI-GL achieves the strongest overall performance on EarthVQA. Under the FT strategy, OA (%), $\kappa(\times 100)$, and mIoU (%) reach 96.01%, 94.81, and 85.80%, respectively. The FZ strategy yields very similar results (96.00%, 94.80, and 85.58%), suggesting that the pretrained encoder transfers effectively to the downstream SPM task on this dataset. Compared with generic SSL baselines (BYOL, SimSiam, and BT), SSI-GL provides the strongest overall results in this experiment and achieves the highest class-wise $F1$-scores on all reported categories. RI and supervised methods still show clear limitations under the 1% label setting. Although increasing the labeled ratio to 10% improves their performance, most supervised results on EarthVQA still remain below the SSL-based results reported in Table III. Overall, these observations suggest that SSL-based pretraining is a promising strategy for low-annotation SPM on this dataset.

For qualitative evaluation, four representative test images are presented in Fig. 9. Since supervised methods cannot produce reliable SPM outputs with only 1% labels, their visual results are shown using models trained with 10% labels. These supervised outputs recover coarse distributions of dominant land-cover regions, but still suffer from blurred boundaries, missing fine structures, and noticeable misclassification artifacts. BYOL, SimSiam, and BT generate cleaner and more coherent maps, yet errors remain in reconstructing thin linear objects and subtle local boundaries. Among the compared methods, SSI-GL appears closest to the GT in both global spatial layout and local detail restoration, with clearer structures and fewer omission or confusion errors.

E. Experiment 3: WHU-H^{2}SR Dataset

Unlike the previous two datasets, WHU-H^{2}SR contains image patches collected from multiple study areas, and patches from the same area may spatially overlap. To avoid potential information leakage caused by random partitioning, we split the dataset by study area rather than by image patch. Specifically, patches from the first 30 study areas were used as the sampling pool for training, whereas patches from the remaining areas were reserved for validation and testing. From

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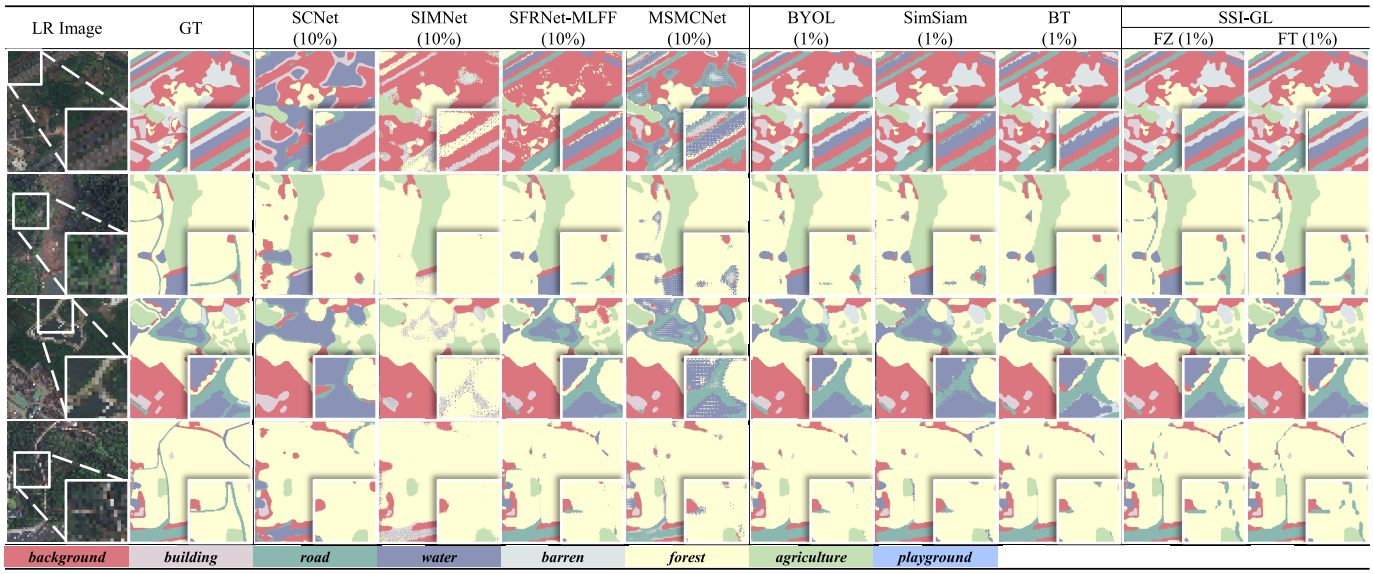


Fig. 9. Visual comparison of SPM results on the EarthVQA dataset.

TABLE IV
QUANTITATIVE COMPARISON ON THE WHU-H^{2}SR DATASET

	SCNet		SIMNet		SFRNet-MLFF		MSMCNet		RI	BYOL	SimSiam	BT	SSI-GL (Ours)	
	1%	10%	1%	10%	1%	10%	1%	10%	1%	1%	1%	1%	FZ (1%)	FT (1%)
OA (%)	43.38	64.43	73.79	91.66	86.45	92.30	44.18	86.77	91.91	95.15	95.28	94.54	**97.77**	**97.93**
$\kappa(\times 100)$	-	44.67	57.63	87.70	78.94	88.49	-	80.38	87.94	92.86	93.06	91.97	**96.72**	**96.97**
mIoU (%)	5.17	13.57	17.45	55.40	35.94	57.82	4.93	38.98	58.19	70.45	74.11	69.72	**86.18**	**87.63**
paddy field	-	0.6252	0.8023	0.9769	0.9498	0.9188	0.0016	0.9804	0.9648	0.9871	0.9836	0.9786	**0.9920**	**0.9921**
dry farmland	0.6350	0.8619	0.8770	0.9712	0.9377	0.9806	0.6047	0.9741	0.9546	0.9780	0.9748	0.9713	**0.9869**	**0.9877**
forest land	-	-	0.0030	0.8981	0.8513	0.9454	-	0.7533	0.8997	0.8946	0.9203	0.9251	**0.9581**	**0.9615**
grassland	-	-	0.0160	0.6392	0.0089	0.9046	-	0.1894	0.7073	0.8791	0.8183	0.8319	**0.9376**	**0.9431**
building	-	-	0.0391	0.8577	0.8226	0.9474	0.0018	0.8244	0.8897	0.9381	0.9031	0.8781	**0.9588**	**0.9643**
highway	-	-	0.0025	0.0955	-	0.0004	-	-	-	0.4880	0.5752	-	**0.7253**	**0.7885**
greenhouse	-	-	0.0035	0.5291	-	-	-	-	-	0.6762	0.6683	0.7144	**0.8356**	**0.8504**
water body	-	-	0.0020	0.2704	0.0013	-	-	-	0.7367	0.6557	0.8749	0.8423	**0.9525**	**0.9506**
background	-	-	0.1457	0.5926	0.0041	0.8628	-	0.2308	0.4858	0.6770	0.7485	0.7607	**0.8892**	**0.8917**

these pools, 1% of the total data was sampled for training, 10% for validation, and the remaining samples were used for testing. All comparative methods followed the same partition protocol. As in the previous experiments, four supervised baselines were additionally trained with 10% labeled data to provide a broader comparison under different supervision levels.

Table IV reports the quantitative results, with SSI-GL achieving the best overall performance on the WHU-H^{2}SR dataset. Under the FT strategy, SSI-GL reaches 97.93% OA, 96.97 for $\kappa(\times 100)$, and 87.63% mIoU. The FZ strategy yields similarly strong results, with 97.77% OA, 96.72 for $\kappa(\times 100)$, and 86.18% mIoU, indicating that the pretrained encoder transfers effectively to the downstream hyperspectral SPM task. RI and the supervised baselines show clear performance limitations under the 1% label setting. Even when the labeled ratio is increased to 10%, the supervised methods still remain below the SSL-based results, and SCNet fails to produce reliable training and testing results under both label ratios. Compared with BYOL, SimSiam, and BT, SSI-GL provides the best overall results in this experiment and maintains clearer advantages on several difficult categories. In particular, classes with fine or sparse spatial structures, such as highway and greenhouse, remain challenging for the supervised baselines, while BYOL and SimSiam still yield relatively low $F1$-scores. Although Barlow Twins improves the reconstruction of greenhouse, it still fails to identify highway effectively. In contrast, SSI-GL achieves the best recognition results for both classes. Overall, these results further demonstrate the advantage of subpixel-aware SSL pretraining for hyperspectral SPM under limited supervision.

For qualitative evaluation, four representative test images are shown in Fig. 10. Consistent with the previous experiments, the supervised methods are visualized using the 10% label setting because their 1% models fail to produce reliable SPM results. SCNet, SIMNet, SFRNet-MLFF, and MSMCNet recover coarse land-cover layouts in some regions, but they generally miss small objects and fine internal structures, resulting in blurred boundaries and fragmented details. BYOL and SimSiam show better recovery of local content, yet their predictions still contain noisy textures and unstable delineation along complex boundaries, leading to visible confusion errors.

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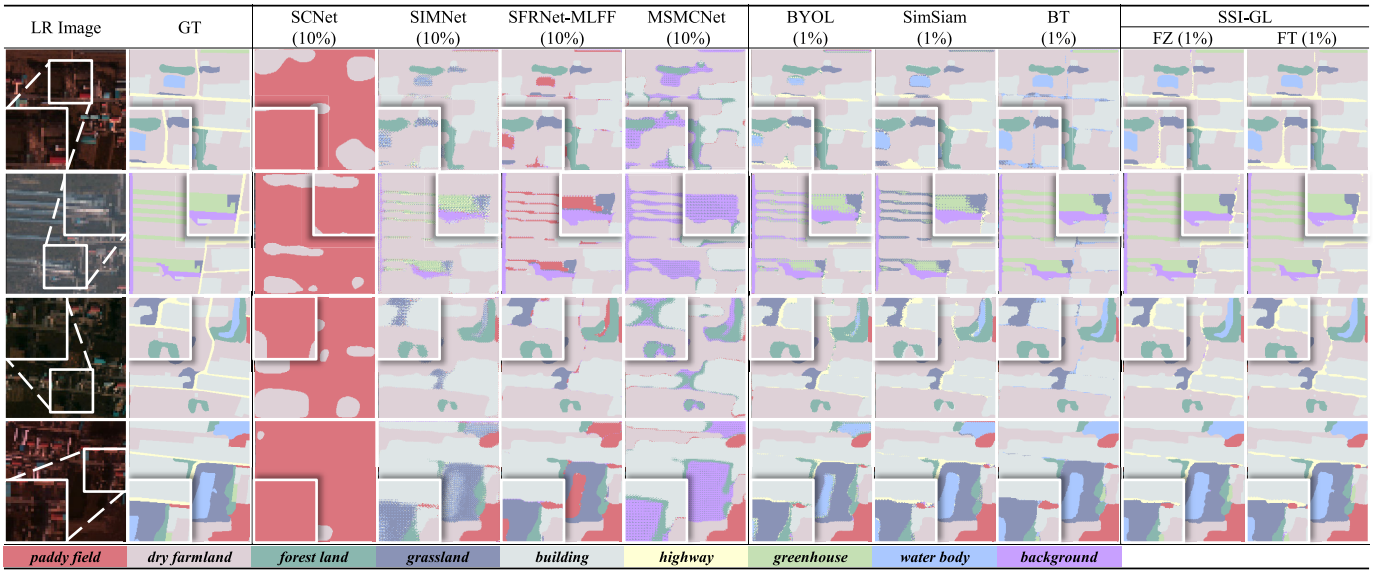


Fig. 10. Visual comparison of SPM results on WHU-H^{2}SR dataset.

TABLE V
QUANTITATIVE COMPARISON ON THE WHU-MT DATASET

	SIMNet	SFRNet-MLFF	MSMCNet	RI	BYOL		SimSiam		BT		SSI-GL (Ours)	
	10%	10%	10%	1%	FZ (1%)	FT (1%)	FZ (1%)	FT (1%)	FZ (1%)	FT (1%)	FZ (1%)	FT (1%)
OA (%)	75.90	81.12	71.31	82.23 ± 0.99	83.91 ± 0.42	83.91 ± 1.41	86.96 ± 0.56	87.43 ± 1.14	86.51 ± 1.08	86.15 ± 1.94	**88.57 ± 0.55**	**88.81 ± 0.53**
κ(×100)	63.15	70.74	53.88	73.54 ± 0.90	76.05 ± 0.68	76.49 ± 2.14	80.44 ± 0.90	81.16 ± 1.79	79.79 ± 1.57	79.22 ± 2.88	**83.01 ± 0.93**	**83.35 ± 0.85**
mIoU (%)	21.71	31.47	17.10	44.98 ± 3.01	43.79 ± 1.42	45.84 ± 3.32	48.04 ± 1.44	49.30 ± 3.12	46.91 ± 1.73	47.97 ± 5.30	**54.24 ± 1.84**	**55.26 ± 1.91**
paddy field	0.8552	0.8975	0.8399	0.8852	0.9018	0.9215	0.9361	0.9409	0.9376	0.9414	**0.9457**	**0.9502**
dry farmland	0.9042	0.8803	0.8606	0.8800	0.9085	0.9015	0.9188	0.9216	0.9172	0.9106	**0.9282**	**0.9255**
forest land	0.0311	0.5499	0.0012	0.7195	0.7166	0.7309	0.7503	0.7657	0.7375	0.6920	**0.7720**	**0.7745**
grassland	0.0010	0.0004	-	0.3290	0.3637	0.4151	0.4648	0.4228	0.4019	0.3703	**0.5827**	**0.6102**
building	0.4670	0.7374	0.0343	0.7602	0.7703	0.7299	0.7655	0.7790	0.7838	0.7761	**0.8037**	**0.7996**
highway	-	-	-	-	-	-	-	-	-	-	-	-
greenhouse	-	-	-	0.5341	0.3299	0.5294	0.3997	0.5584	0.5379	0.6132	**0.6509**	**0.6628**
water body	0.0639	-	-	0.6215	0.5290	0.5231	0.6749	0.6323	0.4434	0.6256	**0.7009**	**0.7358**
background	0.0209	0.3241	0.0434	0.2575	0.3007	0.2888	0.3019	0.2921	0.2739	0.2515	**0.4038**	**0.3993**

BT provides comparatively cleaner results, but omission errors remain for certain subtle categories. In contrast, SSI-GL produces subpixel maps that are visually closest to the GT, with more complete fine-grained spatial patterns, clearer internal structures, and sharper feature boundaries.

F. Experiment 4: WHU-MT Dataset

Given that the soft constraints required by SCNet are unavailable in this dataset and that supervised methods perform poorly with minimal labeled data, our comparative analysis on this more complex real-world dataset was limited to SIMNet, SFRNet-MLFF, and MSMCNet (trained on 10% of the data), together with RI and SSL-based methods (trained on 1%). To implement this experimental design, we followed the data splitting strategy of the WHU-H^{2}SR dataset. The data from the first 30 indexed study areas were partitioned into a training data pool, from which we drew subsets equivalent to 1% and 10% of the total annotated data for training. The remaining study areas were allocated to a combined validation and testing pool. From this latter pool, the validation set was constructed by sampling 10% of the total data volume, with the remainder constituting the test set.

The quantitative evaluation results are presented in Table V. For the real-world WHU-MT benchmark, repeated runs with multiple random seeds were conducted for RI and the SSL-based settings, and the results are reported as mean ± standard deviation to characterize robustness under authentic observation conditions. A comparison between the FZ and FT strategies for SSL-pretrained encoders is also included to further examine their transfer behavior in this real-world setting.

A noticeable performance degradation is observed for all methods on WHU-MT when compared with their results on the simulated datasets. This is expected because the task here is substantially more challenging: the model must recover 1-m HR land cover maps directly from genuine 10-m Sentinel-2 imagery, while also coping with real-data noise and domain complexity. Nevertheless, the repeated-run results show that SSI-GL remains the strongest overall method under the current setting. Under the FT strategy, SSI-GL achieves an OA (%) of 88.81 ± 0.53, a κ(×100) of 83.35 ± 0.85, and an mIoU (%)

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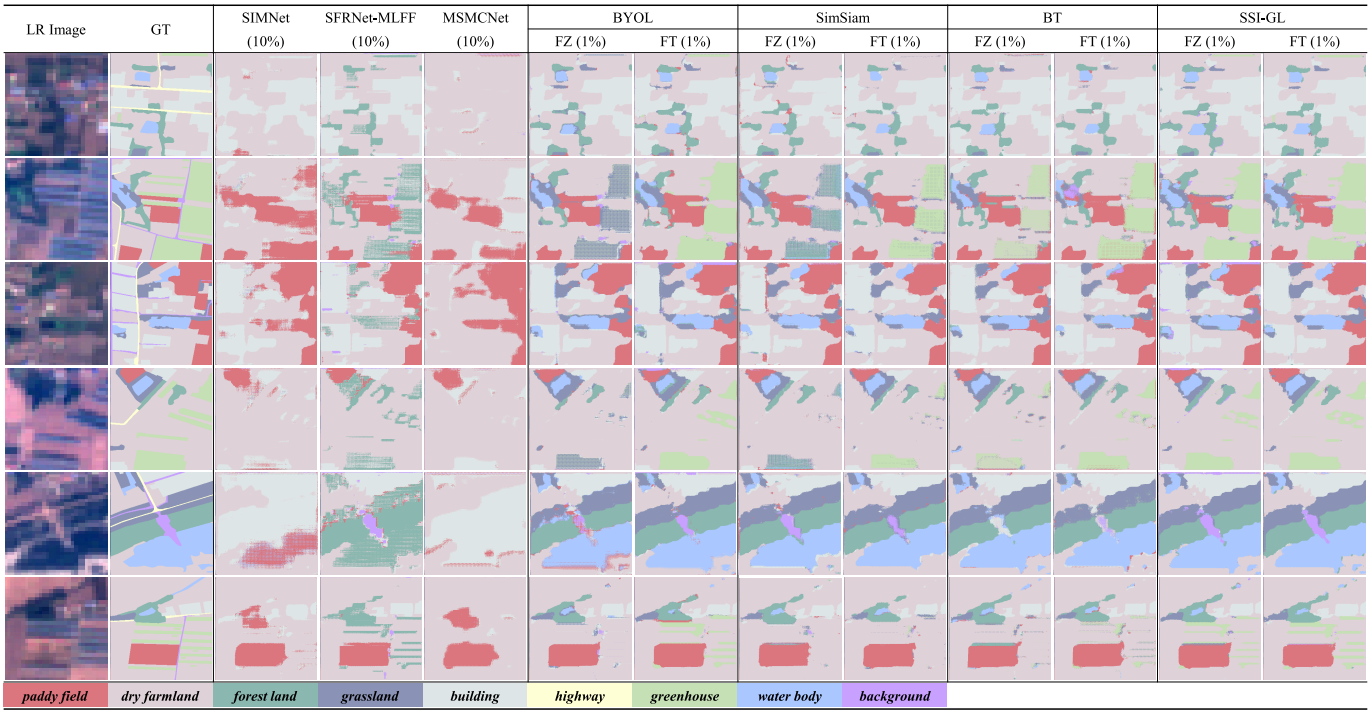


Fig. 11. Visual comparison of SPM results on the WHU-MT dataset.

of 55.26 ± 1.91, while its FZ counterpart also maintains competitive results of 88.57 ± 0.55, 83.01 ± 0.93, and 54.24 ± 1.84, respectively. This indicates that the proposed pretraining transfers effectively even in a more realistic and noisy SPM scenario and that further fine-tuning provides additional gains, especially for class-level reconstruction accuracy. Compared with BYOL, SimSiam, and BT, SSI-GL yields the best mean performance overall and exhibits relatively small fluctuations across repeated runs, suggesting better optimization stability on real-world data. By contrast, SIMNet, SFRNet-MLFF, and MSMCNet, despite being trained with 10% labeled data, still remain clearly below the SSL-based methods and fail to identify multiple land cover categories reliably. These results collectively demonstrate that SSL-based pretraining, particularly the proposed SSI-GL, remains beneficial not only for accuracy but also for robustness on challenging real-data SPM benchmarks. It is also worth noting that none of the methods could effectively identify the "highway" class, highlighting the difficulty of recovering such thin and weakly distinguishable targets from 10-m imagery. To further analyze this issue and compare the visual quality of SPM outputs, six representative test images were selected for qualitative evaluation, as illustrated in Fig. 11.

The visual results are consistent with the quantitative observations. In the LR Sentinel-2 inputs of WHU-MT, the "highway" class contains extremely limited optical and spatial cues, making it difficult for all methods to reconstruct this category effectively. For the remaining categories, SIMNet, SFRNet-MLFF, and MSMCNet mainly recover coarse and continuous land cover regions, but they are highly sensitive to noise and therefore produce many fragmented predictions and speckle-like artifacts. BYOL, SimSiam, and BT reconstruct finer structures than the supervised baselines, yet their outputs still contain noticeable confusion errors and unstable local details. In comparison, SSI-GL better preserves subtle boundaries and intraclass structures of diverse land covers, especially for categories such as paddy field, dry farmland, greenhouse, and water body, and its generated SPM maps are visually closest to the GT in Fig. 11. This qualitative evidence further supports the conclusion that SSI-GL learns more reliable and stable representations for real-world SPM.

G. Encoder Features Visualization

To visually characterize the representation capabilities of various self-supervised pretraining encoders, we applied principal component analysis (PCA) for dimensionality reduction on encoder output features across four datasets and visualized the first two principal components (refer to Fig. 12). Results show that RI yields feature maps characterized by highly blurred, disordered noise textures. The stochastic distribution of high (red) and low (blue) activation regions fails to capture salient structural or edge information from the original LR images, suggesting that encoders without pretraining have limited capacity to extract valid semantic and textural features. The comparison among mainstream self-supervised methods reveals distinct visual characteristics and limitations. BYOL, while demonstrating strong spatial discriminability, suffers from significant high-frequency noise and checkerboard artifacts. Consequently, its local representations appear fragmented, lacking pixel-wise semantic continuity. SimSiam exhibits severe oversmoothing. Although capable of localizing semantic targets at a coarse scale, it significantly compromises high-frequency edge information and texture details,

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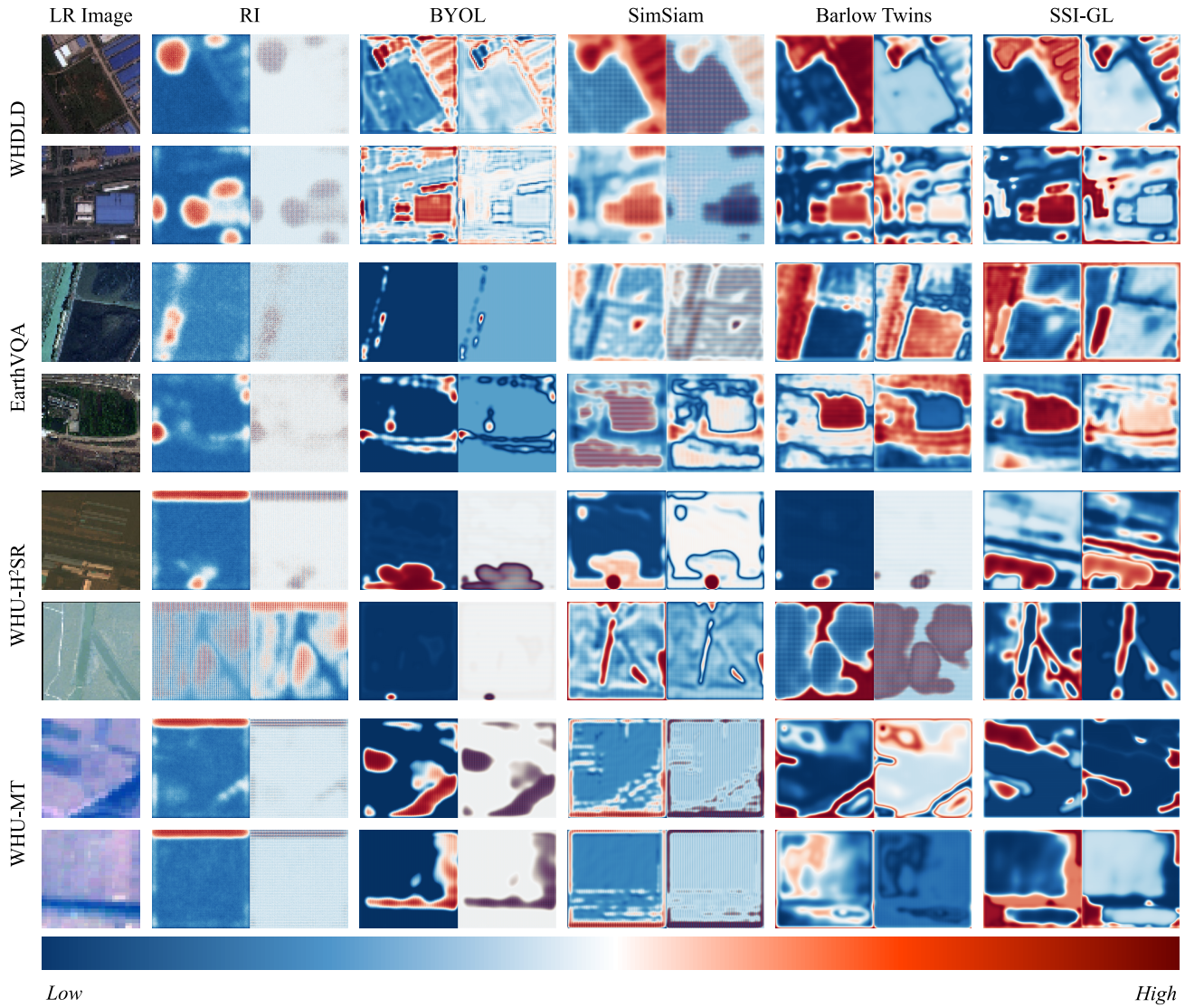


Fig. 12. Visualization of encoder features across the WHDLD, EarthVQA, WHU-H^{2}SR, and WHU-MT datasets.

resulting in indistinct object boundaries. BT enhances edge sharpness to a degree by reducing feature redundancy but remains less effective in capturing complex textures and ensuring intraobject semantic consistency. In contrast, SSI-GL produces feature maps that, in our visualization, show a better balance between structural fidelity and semantic coherence. It generates sharper edges and more distinct geometric structures than BYOL, while retaining richer fine-grained texture information than SimSiam. Particularly in scenarios containing complex terrain features, SSI-GL more clearly distinguishes foreground targets from the background and maintains more uniform activations within objects. These visual comparisons suggest that SSI-GL learns feature representations that are better aligned with the spatial-detail requirements of downstream SPM.

H. Ablation Study

To systematically investigate how attention to subpixel- and image-level features in the SSI-GL framework affects SPM accuracy, we designed an ablation study. Concretely, the global correlation branch and the SSI correlation branch were each selectively retained, and their respective performances were quantitatively evaluated on WHDLD, EarthVQA, WHU-H^{2}SR, and WHU-MT datasets. The results are summarized in Table VI. In this table, the entry for "Global Corr. Loss" directly reports the results of the BT' experiments, as their self-supervised pretraining objectives are equivalent. Under the current experimental setting, the encoder pretrained exclusively with the SSI correlation branch achieves better overall results than the one pretrained solely with the global correlation branch on all four datasets. This suggests that spatially invariant representations at the subpixel level may be more important for the downstream SPM task. Furthermore, removing either branch reduces downstream performance, indicating that both branches contribute useful information to representation learning. While global correlation encodes image-level contextual cues, SSI correlation extracts nuanced local features at the subpixel scale. Taken together, these results support the view that SSI and global correlations play complementary roles in the SSI-GL framework.

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TABLE VI
ABLATION STUDY ON WHDLD, EARTHVQA, WHU-H^{2}SR, AND WHU-MT DATASETS

Datasets	Metrics	RI	Global Corr. Loss		SSI Corr. Loss		SSI-GL (Ours)	
		1%	FZ (1%)	FT (1%)	FZ (1%)	FT (1%)	FZ (1%)	FT (1%)
WHDLD	OA (%)	85.41	94.29	94.25	94.28	94.34	**94.81**	**94.82**
	κ ($\times$100)	78.43	91.78	91.74	91.78	91.87	**92.54**	**92.56**
	mIoU (%)	57.06	84.63	84.41	84.73	84.95	**86.32**	**86.36**
EarthVQA	OA (%)	86.03	94.45	94.77	95.06	94.41	**96.00**	**96.01**
	κ ($\times$100)	81.45	92.76	93.20	93.59	92.74	**94.80**	**94.81**
	mIoU (%)	56.78	76.59	77.12	77.70	76.18	**85.58**	**85.80**
WHU-H^{2}SR	OA (%)	91.91	94.54	94.52	96.26	96.48	**97.77**	**97.93**
	κ ($\times$100)	87.94	91.97	91.87	94.48	94.83	**96.72**	**96.97**
	mIoU (%)	58.19	69.72	69.75	74.68	75.61	**86.18**	**87.63**
WHU-MT	OA (%)	82.23	86.51	86.15	88.20	88.42	**88.57**	**88.81**
	κ ($\times$100)	73.54	79.79	79.22	82.46	82.77	**83.01**	**83.35**
	mIoU (%)	44.98	46.91	47.97	50.71	53.55	**54.24**	**55.26**

TABLE VII
COMPLEXITY COMPARISON IN THE PRETRAINING STAGE

Dataset	Method	Params (M)	Trainable (M)	FLOPs/Batch	Time/Batch
WHDLD	SimSiam	29.9076	29.9076	399.4579 G	297.5643 ms
	BarlowTwins	38.0387	38.0387	399.4905 G	300.7022 ms
	BYOL	59.0269	29.6454	798.9127 G	444.2440 ms
	SSI_GL	42.7654	42.7654	399.5094 G	331.8425 ms
EarthVQA	SimSiam	29.9076	29.9076	399.4579 G	323.8251 ms
	BarlowTwins	38.0387	38.0387	399.4905 G	348.0768 ms
	BYOL	59.0269	29.6454	798.9127 G	492.9096 ms
	SSI_GL	42.7654	42.7654	399.5094 G	499.9130 ms
WHU-H^{2}SR	SimSiam	30.6790	30.6790	409.1351 G	13.4535 s
	BarlowTwins	38.8101	38.8101	409.1676 G	13.4550 s
	BYOL	60.5699	30.4169	818.2670 G	13.3211 s
	SSI_GL	43.5369	43.5369	409.1866 G	13.3458 s
WHU-MT	SimSiam	29.9295	29.9295	399.7333 G	225.0240 ms
	BarlowTwins	38.0606	38.0606	399.7658 G	232.6392 ms
	BYOL	59.0708	29.6674	799.4634 G	343.7188 ms
	SSI_GL	42.7874	42.7874	399.7848 G	242.2396 ms

I. Complexity Discussion

To complement the accuracy analysis, Tables VII and VIII summarize the computational complexity of SSI-GL and the compared methods in the self-supervised pretraining stage and the downstream supervised SPM stage, respectively. We report the numbers of total parameters and trainable parameters, together with the FLOPs and runtime statistics, to provide a clearer comparison of the computational cost and efficiency of different methods.

In the pretraining stage, SSI-GL has a higher parameter count than SimSiam and BT because it adds an SSI-guided aggregation and projection branch on top of the shared encoder. However, its FLOPs per batch remain nearly the same as those of SimSiam and BT on all datasets and are much lower than those of BYOL, whose dual-network design leads to almost doubled FLOPs. This result indicates that the extra computation introduced by SSI-GL is limited mainly to the additional lightweight SSI branch rather than a duplicated encoder forward pass. The measured time per batch does not strictly scale with FLOPs and varies across datasets. For example, SSI-GL is close to BT on WHDLD, WHU-H^{2}SR, and WHU-MT, while a larger runtime gap is observed on EarthVQA despite their similar FLOPs. Overall, Table VII shows that SSI-GL increases pretraining complexity mainly in parameter count while keeping the arithmetic complexity in the same order as the mainstream SSL baselines.

In the downstream stage, SSL-FZ, SSL-RI, and SSL-FT share the same network architecture. Accordingly, their parameter counts and FLOPs remain unchanged within each dataset, which indicates that the performance differences among these settings are associated with the quality of encoder initialization rather than the size of the downstream model. Freezing the pretrained encoder mainly reduces the optimization burden during training, whereas the inference path of the downstream SPM network remains essentially unchanged. Compared with task-specific supervised baselines, the SSL-based SPM framework is still computationally heavier because it relies on a dense high-resolution prediction backbone. However, this cost is

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TABLE VIII
COMPLEXITY AND INFERENCE EFFICIENCY COMPARISON IN THE DOWNSTREAM SUPERVISED STAGE

Dataset	Method	Params (M)	Trainable (M)	FLOPs/Batch	Train/Batch	Infer./Sample
WHDLD	SSL-FZ	34.2694	5.1519	2.4558 T	1.2011 s	40.7636 ms
	SSL-RI	34.2694	34.2694	2.4558 T	1.3575 s	32.2947 ms
	SSL-FT	34.2694	34.2694	2.4558 T	1.3491 s	32.3318 ms
	SIMNet	16.2670	16.2670	148.4642 G	789.6079 ms	8.8346 ms
	SCNet	10.9491	10.9491	116.0611 G	795.4684 ms	5.4689 ms
	SFRNet_MLFF	5.3035	5.3035	114.2669 G	960.6431 ms	15.9542 ms
	MSMCNet	0.1479	0.1479	1.7765 G	672.3818 ms	1.8338 ms
EarthVQA	SSL-FZ	34.2720	5.1544	2.4560 T	1.2556 s	31.7075 ms
	SSL-RI	34.2720	34.2720	2.4560 T	1.4994 s	31.6731 ms
	SSL-FT	34.2720	34.2720	2.4560 T	1.4240 s	31.6770 ms
	SIMNet	16.4216	16.4216	150.0460 G	1.1276 s	8.8456 ms
	SCNet	10.9562	10.9562	116.7170 G	809.1569 ms	5.3255 ms
	SFRNet_MLFF	5.3516	5.3516	115.7678 G	1.0264 s	15.8881 ms
	MSMCNet	0.1485	0.1485	1.7874 G	689.7082 ms	1.8287 ms
WHU-H^2SR	SSL-FZ	35.0447	5.1557	2.4814 T	14.8482 s	33.2035 ms
	SSL-RI	35.0447	35.0447	2.4814 T	14.3867 s	33.4232 ms
	SSL-FT	35.0447	35.0447	2.4814 T	14.5568 s	33.2573 ms
	SIMNet	16.7823	16.7823	153.1620 G	12.9710 s	9.1753 ms
	SCNet	11.2432	11.2432	119.3700 G	13.0288 s	5.6413 ms
	SFRNet_MLFF	5.6597	5.6597	118.8513 G	13.2121 s	16.9198 ms
	MSMCNet	0.2019	0.2019	2.2281 G	12.3190 s	2.0171 ms
WHU-MT	SSL-FZ	34.2952	5.1557	3.3739 T	1.4337 s	45.6880 ms
	SSL-RI	34.2952	34.2952	3.3739 T	1.9041 s	45.7316 ms
	SSL-FT	34.2952	34.2952	3.3739 T	1.8139 s	45.7175 ms
	SIMNet (0.1)	16.5070	16.5070	207.2385 G	754.1675 ms	11.4350 ms
	SFRNet_MLFF (0.1)	5.3843	5.3843	160.1200 G	883.3082 ms	22.0734 ms
	MSMCNet (0.1)	0.1503	0.1503	2.4790 G	784.6782 ms	1.9773 ms

shared by all SSL-based downstream settings and should therefore be attributed to the common SPM architecture instead of being regarded as an extra deployment burden caused by SSI-guided pretraining. Overall, these results suggest that SSI-GL improves downstream SPM mainly through better representation learning while preserving the original inference architecture.

V. CONCLUSION

The continuous acquisition of vast low-to-medium resolution remote sensing imagery offers extensive global coverage; however, the effective utilization of these data is severely constrained by their limited spatial resolution. While SPM provides a viable solution, current supervised DL approaches face a critical bottleneck: the prohibitive labor cost associated with acquiring high-quality, fine-grained HR annotations. To address this challenge, we introduce the SSL paradigm into remote sensing SPM, fundamentally aiming to mitigate the heavy reliance on large-scale LR-HR labeled datasets. In this work, we propose the SSI-GL framework, a novel architecture designed to unlock the potential of SSL for SPM. At the core of our method lies a novel index mask mechanism, which spatially anchors subpixel feature vectors to their relative positions within the original LR images. This design establishes explicit subpixel-level spatial correspondences between augmented views, thereby enhancing the encoder's capability to capture discriminative features during pretraining and reconstruct HR features from LR inputs. Furthermore, we introduce an SCMH for the downstream stage, leveraging LR semantic information to accelerate convergence and refine mapping precision. Across the experiments conducted on four public datasets, SSI-GL achieves the strongest overall results among the compared supervised and SSL methods across key metrics (OA, Kappa, mIoU, and per-class $F1$-scores), particularly in limited-label settings. Moreover, in the real-world application using the WHU-MT dataset, SSI-GL reconstructs 1-m land cover maps from 10-m Sentinel-2 imagery with the best overall quantitative and qualitative results among the compared methods. Overall, these findings suggest that exploring SSL for remote sensing SPM is a promising direction for reducing dependence on costly HR annotations and that SSI-GL provides an effective instantiation of this broader paradigm. Future work will focus on extending SSL-based SPM to more diverse real-world benchmarks and exploring lighter architectures for more efficient deployment.

REFERENCES

[1] H. Liu, K. Wu, D. Zhou, and Y. Xu, "A novel sample generation method for deep learning lithological mapping with airborne TASI hyperspectral data in northern liuyuan, Gansu, China," *Remote Sens.*, vol. 16, no. 15, p. 2852, Aug. 2024.

[2] Y. Dong, Z. Yang, Q. Liu, R. Zuo, and Z. Wang, "Fusion of GaoFen-5 and Sentinel-2B data for lithological mapping using vision transformer dynamic graph convolutional network," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103780.

[3] X. Huang, Y. Dong, Y. Zhang, and B. Du, "Single-source frequency transform for cross-scene classification of hyperspectral image," *IEEE Trans. Image Process.*, vol. 34, pp. 3000–3012, 2025.

[4] W. Tang, K. Wu, Y. Zhang, and Y. Zhan, "A Siamese network based on multiple attention and multilayer transformers for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5219015.

[5] J. Wang, Y. Zhong, A. Ma, Z. Zheng, Y. Wan, and L. Zhang, "LoveNAS: Towards multi-scene land-cover mapping via hierarchical searching adaptive network," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 265–278, Mar. 2024.

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[6] B. Yu, J. Li, and X. Huang, "STSNet: A cross-spatial resolution multimodal remote sensing deep fusion network for high resolution land-cover segmentation," *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102689.

[7] M. Zhao, X. Jiang, and B. Huang, "STFMamba: Spatiotemporal satellite image fusion network based on visual state space model," *ISPRS J. Photogramm. Remote Sens.*, vol. 228, pp. 288–304, Oct. 2025.

[8] D. Zhou, K. Wu, and G. Xu, "A bidirectional cross spatiotemporal fusion network with spectral restoration for remote sensing imagery," *Appl. Sci.*, vol. 15, no. 12, p. 6649, Jun. 2025.

[9] D. He, Q. Shi, J. Xue, P. M. Atkinson, and X. Liu, "Very fine spatial resolution urban land cover mapping using an explicable sub-pixel mapping network based on learnable spatial correlation," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113884.

[10] X. Chen, D. Wang, J. Chen, C. Wang, and M. Shen, "The mixed pixel effect in land surface phenology: A simulation study," *Remote Sens. Environ.*, vol. 211, pp. 338–344, Jun. 2018.

[11] K. Deng, Y. Qian, J. Nie, and J. Zhou, "Diffusion-model-based hyperspectral unmixing using spectral prior distribution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519716.

[12] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[13] T. Wang, J. Li, M. K. Ng, and C. Wang, "Nonnegative matrix functional factorization for hyperspectral unmixing with nonuniform spectral sampling," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401013.

[14] D. He, Y. Zhong, X. Wang, and L. Zhang, "Deep convolutional neural network framework for subpixel mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9518–9539, Nov. 2021.

[15] J. Zhong, K. Wu, and Y. Xu, "Deep spatial feedback refined network with multilevel feature fusion for hyperspectral image subpixel mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409620.

[16] P. M. Atkinson, M. E. J. Cutler, and H. Lewis, "Mapping sub-pixel proportional land cover with AVHRR imagery," *Int. J. Remote Sens.*, vol. 18, no. 4, pp. 917–935, Mar. 1997.

[17] D. He, Q. Shi, X. Liu, Y. Zhong, and X. Zhang, "Deep subpixel mapping based on semantic information modulated network for urban land use mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10628–10646, Dec. 2021.

[18] A. J. Tatem, H. G. Lewis, P. M. Atkinson, and M. S. Nixon, "Super-resolution target identification from remotely sensed images using a Hopfield neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 4, pp. 781–796, Apr. 2001.

[19] P. M. Atkinson, "Sub-pixel target mapping from soft-classified, remotely sensed imagery," *Photogramm. Eng. Remote Sens.*, vol. 71, no. 7, pp. 839–846, Jul. 2005.

[20] K. C. Mertens, B. De Baets, L. P. Verbeke, and R. R. De Wulf, "A sub-pixel mapping algorithm based on sub-pixel/pixel spatial attraction models," *Int. J. Remote Sens.*, vol. 27, no. 15, pp. 3293–3310, 2006.

[21] L. Wang, Q. Wang, and D. Liu, "Sub-pixel mapping based on sub-pixel to sub-pixel spatial attraction model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 593–596.

[22] F. Ling, G. M. Foody, Y. Ge, X. Li, and Y. Du, "An iterative interpolation deconvolution algorithm for superresolution land cover mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7210–7222, Dec. 2016.

[23] P. Wang, Y. Wu, and H. Leung, "Subpixel land cover mapping based on a new spatial attraction model with spatial–spectral information," *Int. J. Remote Sens.*, vol. 40, no. 16, pp. 6444–6463, Aug. 2019.

[24] Y. Zhong, Y. Wu, L. Zhang, and X. Xu, "Adaptive MAP sub-pixel mapping model based on regularization curve for multiple shifted hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 96, pp. 134–148, Oct. 2014.

[25] Y. Zhong, Y. Wu, X. Xu, and L. Zhang, "An adaptive subpixel mapping method based on MAP model and class determination strategy for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1411–1426, Mar. 2015.

[26] P. Wang, L. Wang, and J. Chanussot, "Soft-then-hard subpixel land cover mapping based on spatial–spectral interpolation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1851–1854, Dec. 2016.

[27] K. Wu, Q. Du, X. Hu, and X. Wang, "Sub-pixel mapping based on MAP model and spatial attraction theory for remotely sensed image," *IEEE Access*, vol. 5, pp. 25126–25132, 2017.

[28] M. Song, Y. Zhong, A. Ma, and R. Feng, "Multiobjective sparse subpixel mapping for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4490–4508, Jul. 2019.

[29] Y. Chen et al., "Subpixel mapping for remote sensing imagery based on spatial adaptive attraction model and conditional random fields," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1624–1640, 2023.

[30] X. Li, F. Ling, G. M. Foody, Y. Ge, Y. Zhang, and Y. Du, "Generating a series of fine spatial and temporal resolution land cover maps by fusing coarse spatial resolution remotely sensed images and fine spatial resolution land cover maps," *Remote Sens. Environ.*, vol. 196, pp. 293–311, Jul. 2017.

[31] D. He, Y. Zhong, and L. Zhang, "Spatiotemporal subpixel geographical evolution mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2198–2220, Apr. 2019.

[32] R. Chen et al., "Spatiotemporal continuous impervious surface mapping by fusion of Landsat time series data and Google Earth imagery," *Remote Sens.*, vol. 13, no. 12, p. 2409, Jun. 2021.

[33] K. C. Mertens, L. P. Verbeke, and R. Wulf, "Sub-pixel mapping with neural networks: Real-world spatial configurations learned from artificial shapes," in *Proc. 4th Int. Symp. Remote Sens. Urban Areas*, 2003, pp. 117–121.

[34] L. Zhang, K. Wu, Y. Zhong, and P. Li, "A new sub-pixel mapping algorithm based on a BP neural network with an observation model," *Neurocomputing*, vol. 71, nos. 10–12, pp. 2046–2054, Jun. 2008.

[35] Y. Zhang, P. M. Atkinson, X. Li, F. Ling, Q. Wang, and Y. Du, "Learning-based spatial–temporal superresolution mapping of forest cover with MODIS images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 600–614, Jan. 2017.

[36] P. V. Arun, K. M. Buddhiraju, and A. Porwal, "CNN based sub-pixel mapping for hyperspectral images," *Neurocomputing*, vol. 311, pp. 51–64, Oct. 2018.

[37] Y. Jia, Y. Ge, Y. Chen, S. Li, G. B. M. Heuvelink, and F. Ling, "Super-resolution land cover mapping based on the convolutional neural network," *Remote Sens.*, vol. 11, no. 15, p. 1815, Aug. 2019.

[38] X. Ma, Y. Hong, Y. Song, and Y. Chen, "A super-resolution convolutional-neural-network-based approach for subpixel mapping of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4930–4939, Dec. 2019.

[39] X. Zhang, Y. Ge, F. Ling, J. Chen, Y. Chen, and Y. Jia, "Graph convolutional networks-based super-resolution land cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7667–7681, 2021.

[40] F. Ling and G. M. Foody, "Super-resolution land cover mapping by deep learning," *Remote Sens. Lett.*, vol. 10, no. 6, pp. 598–606, Jun. 2019.

[41] X. Zhang et al., "High-quality super-resolution mapping using spatial deep learning," *iScience*, vol. 26, no. 6, Jun. 2023, Art. no. 106875.

[42] M. Xu, X. Zou, S. Liu, H. Sheng, and Y. Dong, "Multiscale semantically modulated mixed convolutional networks for subpixel mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5521816.

[43] M. Grau, A. Lontke, X. Jiang, and L. Scheibenreif, "Self supervised learning in remote sensing: Quantifying approaches effectiveness across downstream tasks," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 518–521.

[44] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[45] V. Rani, S. T. Nabi, M. Kumar, A. Mittal, and K. Kumar, "Self-supervised learning: A succinct review," *Arch. Comput. Methods Eng.*, vol. 30, no. 4, pp. 2761–2775, May 2023.

[46] M. C. Schiappa, Y. S. Rawat, and M. Shah, "Self-supervised learning for videos: A survey," *ACM Comput. Surv.*, vol. 55, no. 13s, pp. 1–37, Dec. 2023.

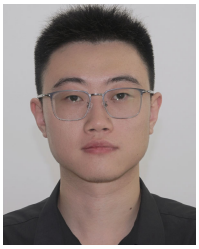
[47] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.

[48] Y. Liu, J. Zeng, X. Tao, and G. Fang, "Rethinking self-supervised semantic segmentation: Achieving end-to-end segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10036–10046, Dec. 2024.

[49] Z.-Y. Li, B.-W. Yin, Y. Liu, L. Liu, and M.-M. Cheng, "Enhancing representations through heterogeneous self-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5976–5989, Jul. 2025.

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[50] M. Nie, X. Cai, H. Xu, and L. Zhang, "LaneCorrect: Self-supervised lane detection," *Int. J. Comput. Vis.*, vol. 133, no. 7, pp. 4894–4908, Jul. 2025.
[51] Q. Zhu, Z. Li, T. Song, L. Yao, Q. Guan, and L. Zhang, "Unrestricted region and scale: Deep self-supervised building mapping framework across different cities from five continents," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 344–367, Mar. 2024.
[52] Y. Chen, F. Huang, and L. Cao, "Aperture array-based self-supervised hyperspectral super-resolution imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5526117.
[53] H. Xu, C. Zhang, P. Yue, and K. Wang, "SDCluster: A clustering based self-supervised pre-training method for semantic segmentation of remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 223, pp. 1–14, May 2025.
[54] R. Yang, Y. Zhong, and Y. Su, "Self-supervised joint representation learning for urban land-use classification with multisource geographic data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5608021.
[55] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
[56] J.-B. Grill et al., "Bootstrap your own latent—A new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 21271–21284.
[57] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 15750–15758.
[58] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12310–12320.
[59] X. Wang, R. Zhang, C. Shen, T. Kong, and L. Li, "Dense contrastive learning for self-supervised visual pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3024–3033.
[60] Z. Xie, Y. Lin, Z. Zhang, Y. Cao, S. Lin, and H. Hu, "Propagate yourself: Exploring pixel-level consistency for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16679–16688.
[61] D. Muhtar, X. Zhang, and P. Xiao, "Index your position: A novel self-supervised learning method for remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4411511.
[62] Z. Shao, W. Zhou, X. Deng, M. Zhang, and Q. Cheng, "Multilabel remote sensing image retrieval based on fully convolutional network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 318–328, 2020.
[63] J. Wang, Z. Zheng, Z. Chen, A. Ma, and Y. Zhong, "EarthVQA: Towards queryable Earth via relational reasoning-based remote sensing visual question answering," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 5481–5489.



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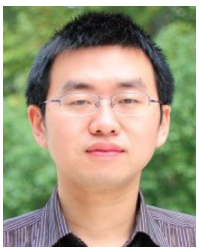
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